



```
import pandas as pd
import matplotlib.pyplot as plt
data = {'Year': list(range(2010, 2021)),
        'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}

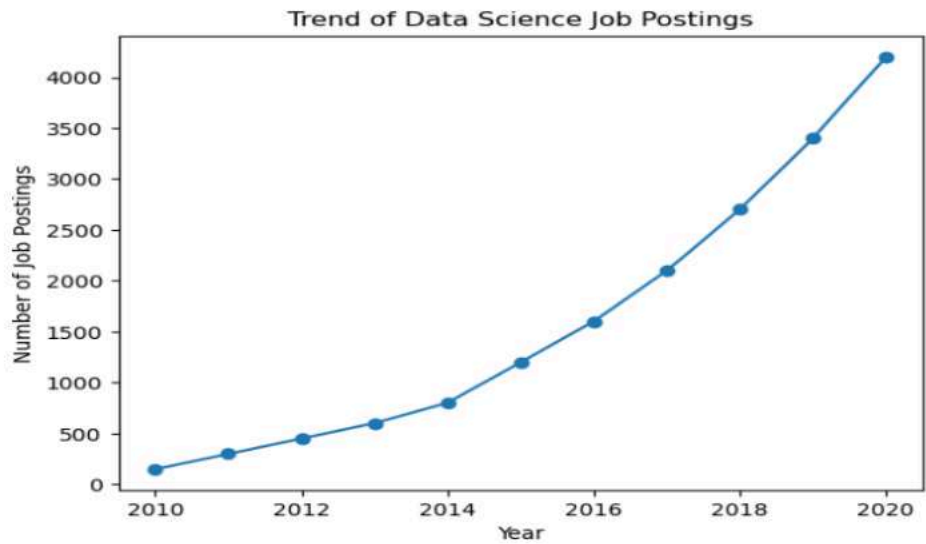
df = pd.DataFrame(data)
plt.plot(df['Year'], df['Job Postings'], marker='o')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number of Job Postings')
plt.show()
```

Sample Data Input:

Year =2010, 2021

Job Postings=150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200

Sample Output:





```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer', 'Business Analyst']
counts = [300, 500, 450, 200, 150]
plt.bar(roles, counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.show()
```

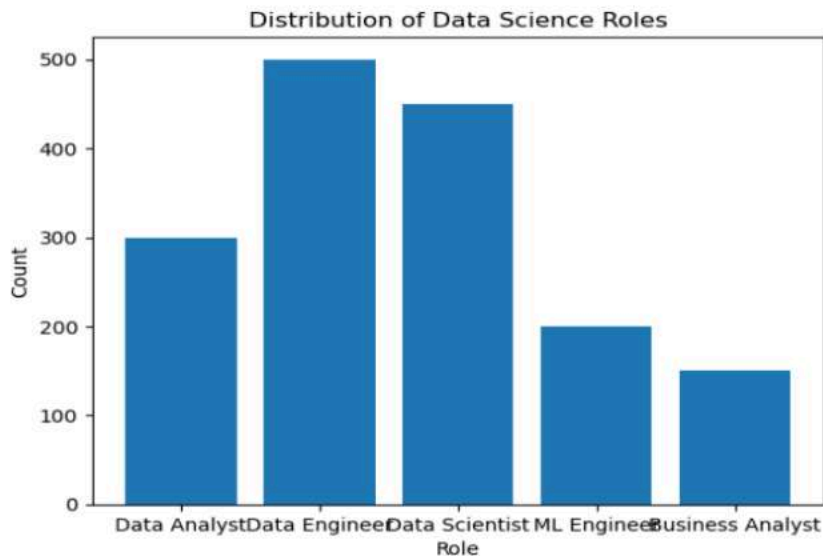
✓ 0.2s

Sample Data Input:

roles = Data Analyst, Data Engineer, Data Scientist, ML Engineer, Business Analyst.

counts = 300, 500, 450, 200, 150.

Sample Output:





```
# Structured data example
structured_data = pd.DataFrame({
    'ID': [1, 2, 3],
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35]
})
print("Structured Data:\n", structured_data)

# Unstructured data example
unstructured_data = "This is an example of unstructured data. It can be a piece of text, an image, or
a video file."
print("\nUnstructured Data:\n", unstructured_data)

# Semi-structured data example (JSON)
semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
print("\nSemi-structured Data:\n", semi_structured_data)
```

✓ 0.1s

OUTPUT:

Structured Data:

	ID	Name	Age
0	1	Alice	25
1	2	Bob	30
2	3	Charlie	35

Unstructured Data:

This is an example of unstructured data. It can be a piece of text, an image, or a video file.

Semi-structured Data:

```
{'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
}
```



Generate key and encrypt data

```
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
```

✓ 0.1s

b'...'



```
f.decrypt(token)
```

✓ 0.1s

b'Rajalakshmi Engineering College'



```
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)

# Decrypt data
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

✓ 0.1s

OUTPUT:

```
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data: b'gAAAAABmmnHEhy1ZC1-Dox-URNAvYqFFGfJemMb_fKmXqhATe3AAAn
XEOOnSzaNRINI4zAbNK8lgx2pprrNn-j6agbWDF30OwbrZsBuz8fnDdX7N35WwloDu3m6pVv
kwv8GYDSI8FovltV'
Decrypted Data: b'Rajalakshmi Engineering College.'
```



```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
# Load the data into a pandas DataFrame
file_path = 'C:\sales_data.csv'
df = pd.read_csv(file_path)
# Display the first few rows of the DataFrame
print(df.head())
# Check for missing values
print(df.isnull().sum())
# Fill or drop missing values if necessary
df['Sales'].fillna(df['Sales'].mean(), inplace=True)
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
# Summary statistics
print(df.describe())
# Group by product and calculate the total sales and quantity
product_summary = df.groupby('Product').agg({
    'Sales': 'sum',
    'Quantity': 'sum'
}).reset_index()
print(product_summary)
# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
# Line plot of sales over time
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date').agg({'Sales': 'sum'}).reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'], sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('Sales Over Time')
plt.show()
# Pivot table to analyze sales by region and product
pivot_table = df.pivot_table(values='Sales', index='Region', columns='Product',
                               aggfunc=np.sum, fill_value=0)
print(pivot_table)
# Correlation matrix
correlation_matrix = df.corr()
print(correlation_matrix)
# Heatmap of the correlation matrix
import seaborn as sns
plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()
```

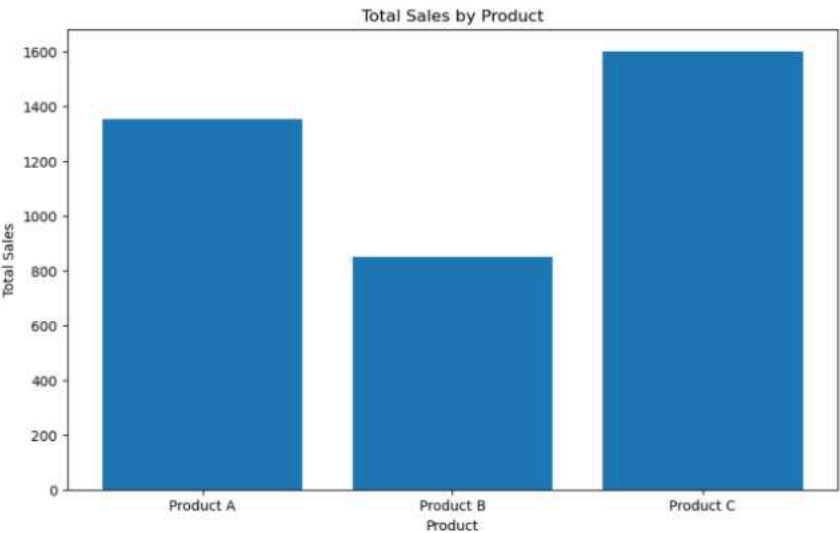
OUTPUT:

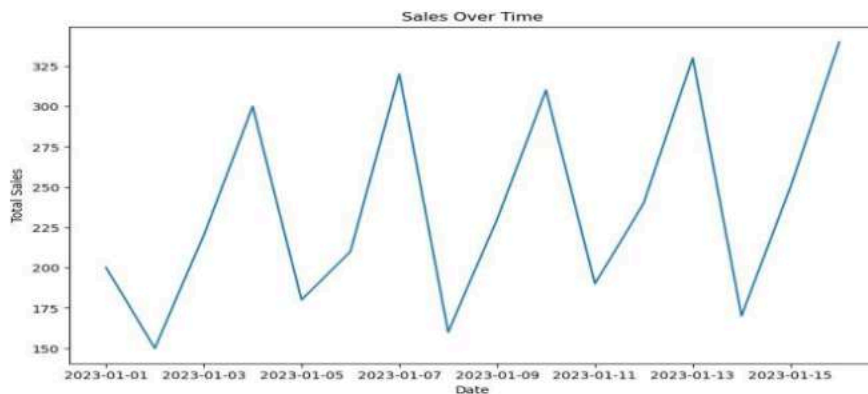
	Date	Product	Sales	Quantity	Region
0	2023-01-01	Product A	200	4	North
1	2023-01-02	Product B	150	3	South
2	2023-01-03	Product A	220	5	North
3	2023-01-04	Product C	300	6	East
4	2023-01-05	Product B	180	4	West

```
Date      0
Product    0
Sales      0
Quantity   0
Region     0
dtype: int64
```

	Sales	Quantity
count	16.000000	16.000000
mean	237.500000	5.375000
std	64.031242	1.746425
min	150.000000	3.000000
25%	187.500000	4.000000
50%	225.000000	5.500000
75%	302.500000	7.000000
max	340.000000	8.000000

	Product	Sales	Quantity
0	Product A	1350	33
1	Product B	850	17
2	Product C	1600	36



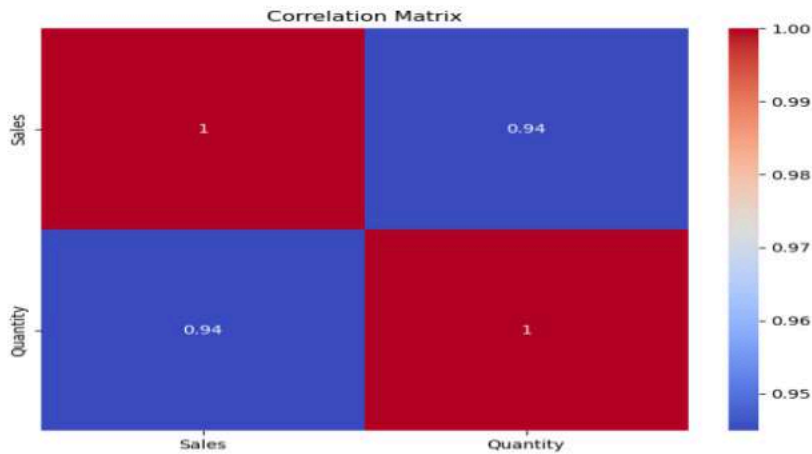


Product	Product A	Product B	Product C
Region			
East	0	0	1600
North	1350	0	0
South	0	480	0
West	0	370	0

	Sales	Quantity
Sales	1.000000	0.944922
Quantity	0.944922	1.000000

C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_9648\511106317.py:49: FutureWarning: The default value of numeric_only in DataFrame.corr is deprecated . In a future version, it will default to False. Select only valid columns or specify the value of numeric_only to silence this warning.

```
correlation_matrix = df.corr()
```

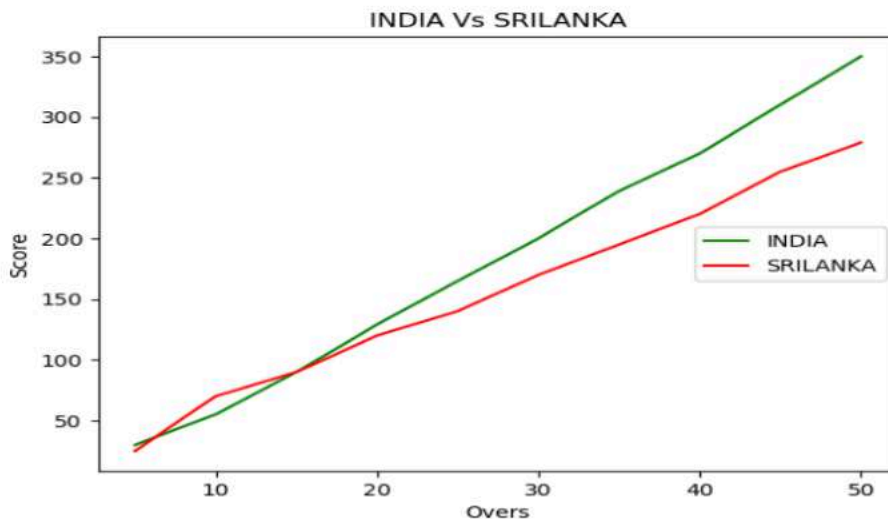




```
import matplotlib.pyplot as cricket
Overs = list(range(5, 51, 5))
Indian_Score = [30, 55, 90, 129, 165, 200, 239, 270, 310, 350]
Srilankan_Score = [25, 70, 90, 120, 140, 170, 195, 220, 255, 279]
cricket.plot(Overs, Indian_Score)
cricket.plot(Overs, Srilankan_Score)
cricket.show()
cricket.title("INDIA Vs SRILANKA")
cricket.xlabel("Overs")
cricket.ylabel("Score")
cricket.legend()
cricket.plot(Overs, Indian_Score, color="green", label="INDIA")
cricket.plot(Overs, Srilankan_Score, color="red", label="SRILANKA")
cricket.legend(loc="center right")
```

✓ 0.4s

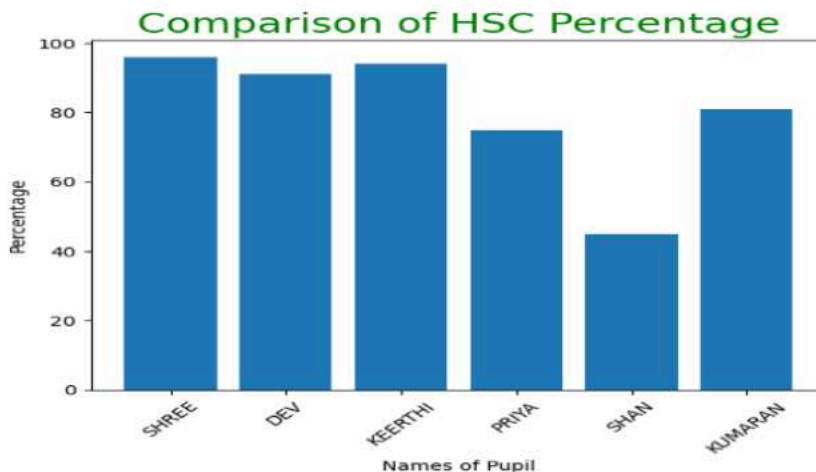
OUTPUT:





```
import matplotlib.pyplot as hscmark
import numpy as np
Names = ['SHREE', 'DEV', 'KEERTHI', 'PRIYA', 'SHAN', 'KUMARAN']
xaxis = np.arange(len(Names))
Percentage_hsc = [96, 91, 94, 75, 45, 81]
hscmark.bar(Names, Percentage_hsc)
hscmark.xticks(xaxis, Names, rotation=45)
hscmark.xlabel("Names of Pupil")
hscmark.ylabel("Percentage")
hscmark.title("Comparison of HSC Percentage", fontsize=20, color="green")
hscmark.show()
```

✓ 0.3s

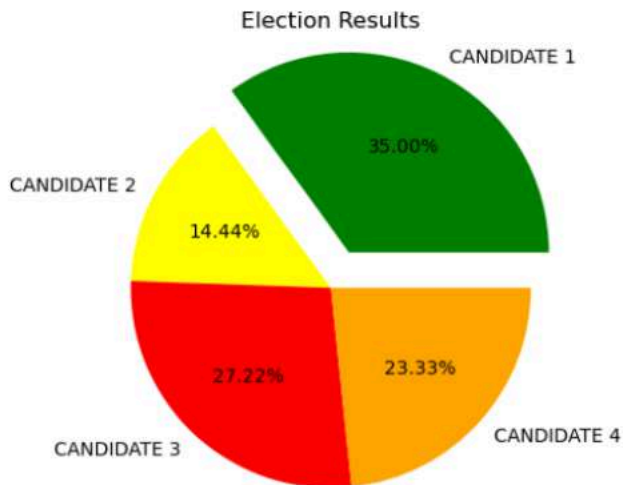
OUTPUT:



```
import matplotlib.pyplot as election
# Election data
labels = ['CANDIDATE 1', 'CANDIDATE 2', 'CANDIDATE 3', 'CANDIDATE 4']
Votes = [315, 130, 245, 210]
colors = ['green', 'yellow', 'red', 'orange']
explode = (0.2, 0, 0, 0)
# Plotting the pie chart
election.pie(Votes, labels=labels, colors=colors, explode=explode, autopct='%0.2f%%')
election.title('Election Results')
election.show()
```

✓ 0.3s

OUTPUT:





```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import gutenberg
# Download the gutenberg corpus if not already installed
nltk.download('gutenberg')
nltk.download('punkt') # Ensure the punkt tokenizer models are also downloaded
# Load the text from the gutenberg corpus
sample = gutenberg.raw("austen-emma.txt") # Change to an existing text from the corpus
# Tokenize the sample text
token = word_tokenize(sample)
# Create a list of the first 50 tokens
wlist = []
for i in range(50):
    wlist.append(token[i])
# Calculate the frequency of each word in the list
wordfreq = [wlist.count(w) for w in wlist]
# Print the word-frequency pairs
print("Pairs\n" + str(list(zip(wlist, wordfreq))))
```

✓ 1.9s

OUTPUT:

Pairs

```
[(['', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816',
1), (',', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma',
2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever',
1), (',', 5), ('and', 3), ('rich', 1), (',', 5), ('with', 2), ('a', 1),
('comfortable', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposi
tion', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1),
('of', 2), ('the', 2), ('best', 1), ('blessings', 1), ('of', 2), ('e
xistence', 1), (',', 1), ('and', 3), ('had', 1), ('lived', 1), ('nearly',
1), ('twenty-one', 1), ('years', 1), ('in', 1), ('the', 2), ('world',
1), ('with', 2)]
```

In []:

```
import pandas as pd

# Load the dataset
diabetes_df = pd.read_csv("diabetes.csv")

# Display the first few rows
print(diabetes_df.head())
```

```
[1] import pandas as pd
db= pd.read_csv("/content/drive/MyDrive/PIMA Indian Dataset.csv")
```

```
print(db.head())
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	
	DiabetesPedigreeFunction		Age	Outcome			
0	0.627		50	1			
1	0.351		31	0			
2	0.672		32	1			
3	0.167		21	0			
4	2.288		33	1			

2. Initial Exploration:

```

▶ # Dataset structure and summary statistics
print(db.info())
print(db.describe())

# Visualize the distribution of each feature
import matplotlib.pyplot as plt
import seaborn as sns

db.hist(bins=50, figsize=(20, 15))
plt.show()

# Pairplot to visualize relationships
sns.pairplot(db)
plt.show()

```

[3]

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 768 entries, 0 to 767
```

```
Data columns (total 9 columns):
```

#	Column	Non-Null Count	Dtype
0	Pregnancies	768 non-null	int64
1	Glucose	768 non-null	int64
2	BloodPressure	768 non-null	int64
3	SkinThickness	768 non-null	int64
4	Insulin	768 non-null	int64
5	BMI	768 non-null	float64
6	DiabetesPedigreeFunction	768 non-null	float64
7	Age	768 non-null	int64
8	Outcome	768 non-null	int64

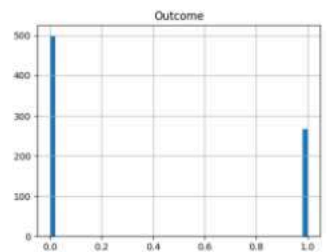
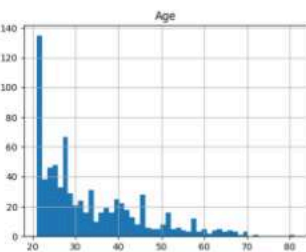
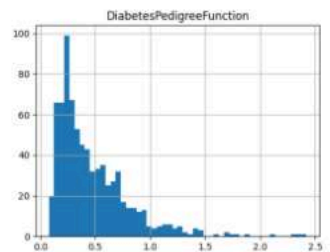
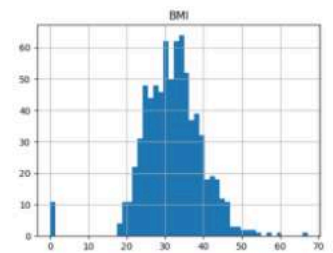
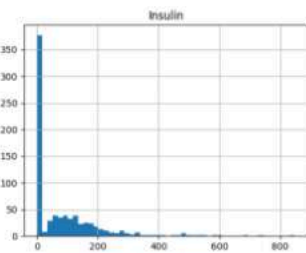
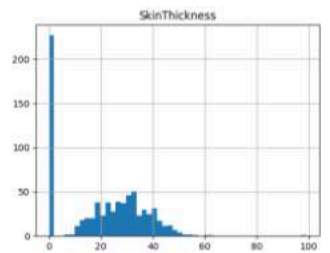
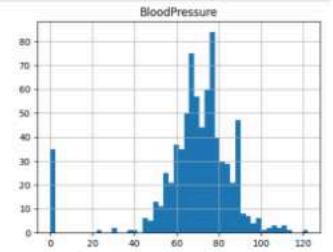
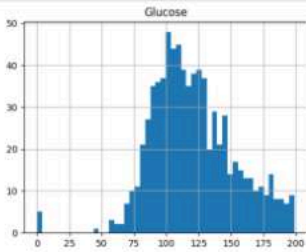
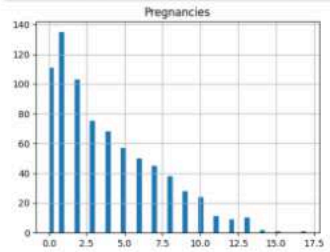
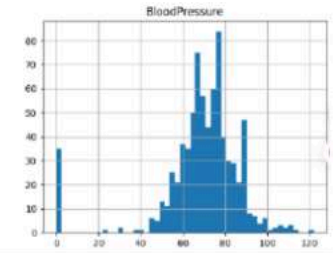
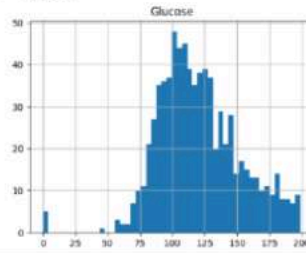
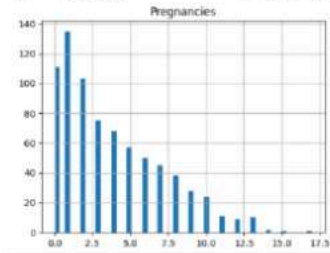
```
dtypes: float64(2), int64(7)
```

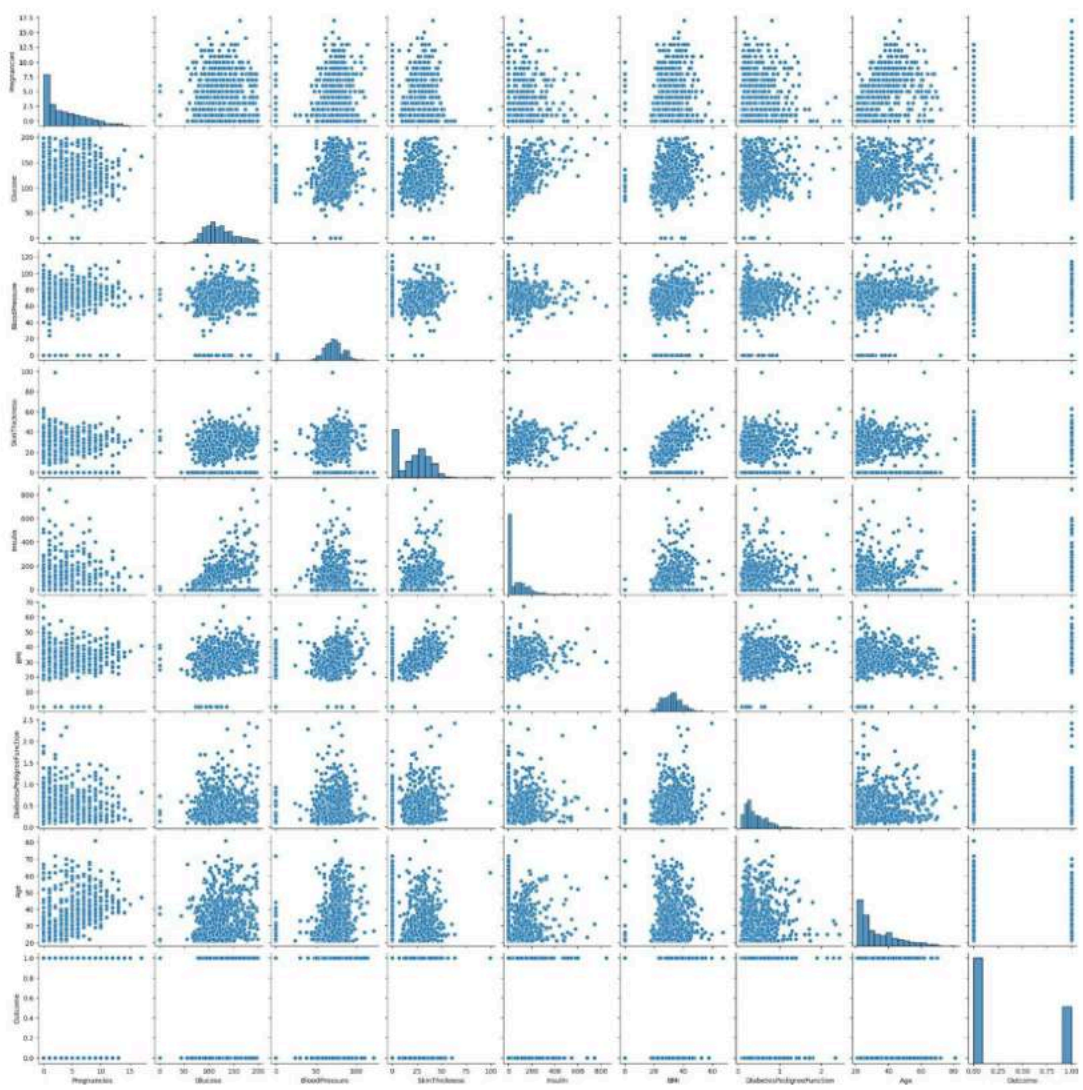
```
memory usage: 54.1 KB
```

```
None
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin \
count	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479
std	3.369578	31.972618	19.355807	15.952218	115.244002
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000
75%	6.000000	140.250000	80.000000	32.000000	127.250000
max	17.000000	199.000000	122.000000	99.000000	846.000000

	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	31.992578	0.471876	33.240885	0.348958
std	7.894196	0.331329	11.760232	0.476951
min	0.000000	0.070000	21.000000	0.000000
25%	27.300000	0.243750	24.000000	0.000000
50%	31.000000	0.372500	29.000000	0.000000
75%	36.000000	0.626250	41.000000	1.000000
max	67.100000	2.420000	81.000000	1.000000





In []:

```
import pandas as pd

import matplotlib.pyplot as plt

from statsmodels.tsa.seasonal import seasonal_decompose

# Load dataset

data = pd.read_csv('website_traffic.csv', parse_dates=['Date'], index_col='Date')

# Time Series Decomposition

decomposition = seasonal_decompose(data['Page_Views'], model='additive')

decomposition.plot()
```

```
plt.show()
```

```
# Moving Average
```

```
data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()
```

```
data[['Page_Views', 'Page_Views_MA']].plot()
```

```
plt.title('Page Views with Moving Average')
```

```
plt.show()
```

```
# Anomaly Detection
```

```
data['Page_Views_Z'] = (data['Page_Views'] - data['Page_Views'].mean()) /  
data['Page_Views'].std()
```

```
anomalies = data[data['Page_Views_Z'].abs() > 3]
```

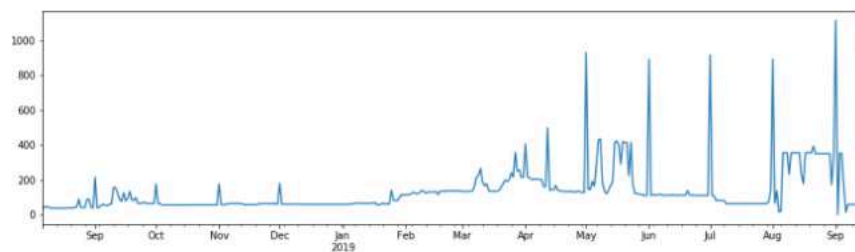
```
plt.plot(data.index, data['Page_Views'], label='Page Views')
```

```
plt.scatter(anomalies.index, anomalies['Page_Views'], color='red', label='Anomalies')
```

```
plt.legend()
```

```
plt.show()
```

sample output



Output

In []:

Copy code

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
# Step 1: Generate a population (e.g., normal distribution)
```

```
population_mean = 50
```

```
population_std = 10
```

```
population_size = 100000
```

```
population = np.random.normal(population_mean, population_std, population_size)
```

```
# Step 2: Random sampling
```

```
sample_sizes = [30, 50, 100] # different sample sizes to consider
```

```
num_samples = 1000 # number of samples for each sample size
```

```

sample_means = {}

for size in sample_sizes:
    sample_means[size] = []
    for _ in range(num_samples):
        sample = np.random.choice(population, size=size, replace=False)
        sample_means[size].append(np.mean(sample))

# Step 3: Plotting sampling distributions
plt.figure(figsize=(12, 8))

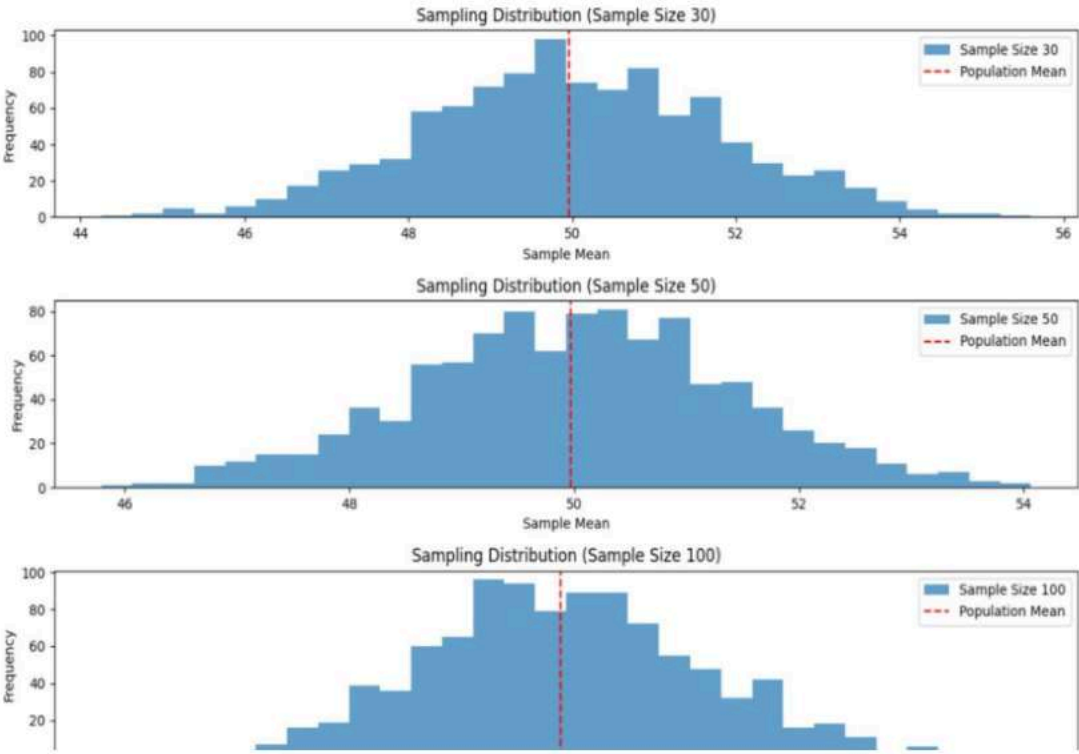
for i, size in enumerate(sample_sizes):
    plt.subplot(len(sample_sizes), 1, i+1)
    plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size {size}')
    plt.axvline(np.mean(population), color='red', linestyle='dashed', linewidth=1.5, label='Population
Mean')

    plt.title(f'Sampling Distribution (Sample Size {size})')
    plt.xlabel('Sample Mean')
    plt.ylabel('Frequency')
    plt.legend()

plt.tight_layout()
plt.show()

```

OUTPUT:



In []:

```
import numpy as np

import scipy.stats as stats


# Define the sample data (hypothetical weights in grams)

sample_data = np.array([152, 148, 151, 149, 147, 153, 150, 148, 152, 149, 151, 150, 149, 152, 151,
148, 150, 152, 149, 150, 148, 153, 151, 150, 149, 152, 148, 151, 150, 153])

# Population mean under the null hypothesis

population_mean = 150

# Calculate sample statistics

sample_mean = np.mean(sample_data)

sample_std = np.std(sample_data, ddof=1) # Using sample standard deviation


# Number of observations

n = len(sample_data)

# Calculate the Z-statistic

z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))


# Calculate the p-value
```

```
p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic))) # Two-tailed test

# Print results

print(f"Sample Mean: {sample_mean:.2f}")
print(f"Z-Statistic: {z_statistic:.4f}")
print(f"P-Value: {p_value:.4f}")

# Decision based on the significance level

alpha = 0.05

if p_value < alpha:
    print("Reject the null hypothesis: The average weight is significantly different from 150 grams.")
else:
    print("Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.")
```

OUTPUT:

Sample Mean: 150.20

Z-Statistic: 0.6406

P-Value: 0.5218

Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.

In []:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
```

```
data=pd.read_csv('/content/Iris_Dataset.csv')
data
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	variety
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column              Non-Null Count  Dtype
---  -
0    Id                   150 non-null   int64
1    SepalLengthCm       150 non-null   float64
2    SepalWidthCm        150 non-null   float64
3    PetalLengthCm       150 non-null   float64
4    PetalWidthCm        150 non-null   float64
5    variety              150 non-null   object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```
data.describe()
```

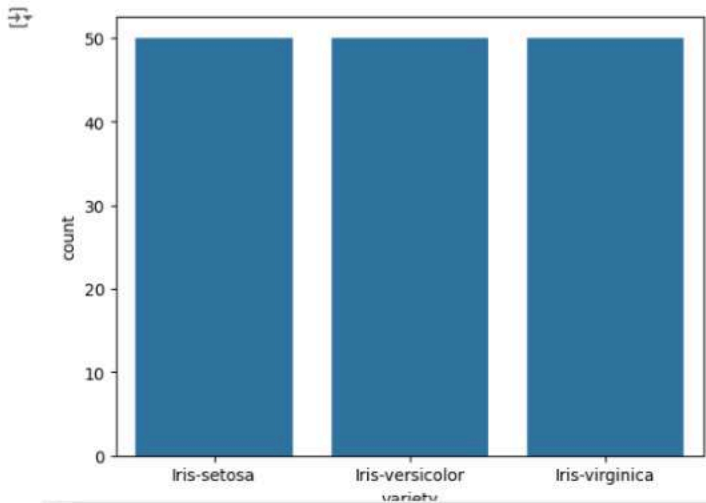
	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
data.value_counts('variety')
```

	count
variety	
Iris-setosa	50
Iris-versicolor	50
Iris-virginica	50

In []:

```
sns.countplot(x='variety',data=data,)
plt.show()
```



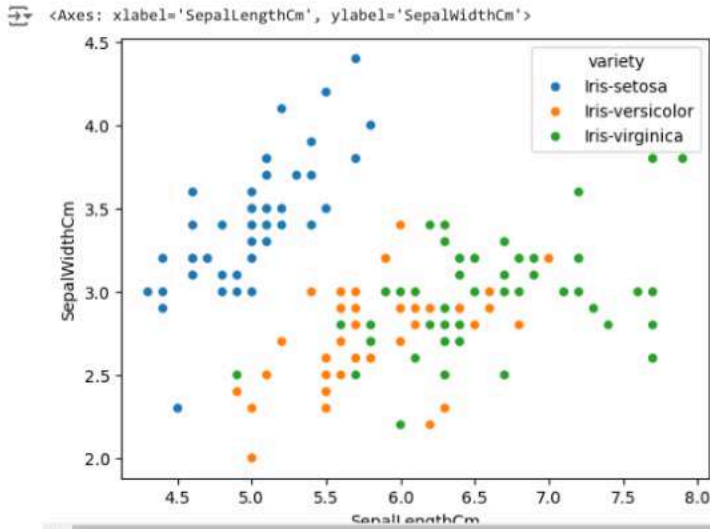
```
dummies=pd.get_dummies(data.variety)

FinalDataset=pd.concat([pd.get_dummies(data.variety),data.iloc[:,[0,1,2,3]]],axis=1)

FinalDataset.head()
```

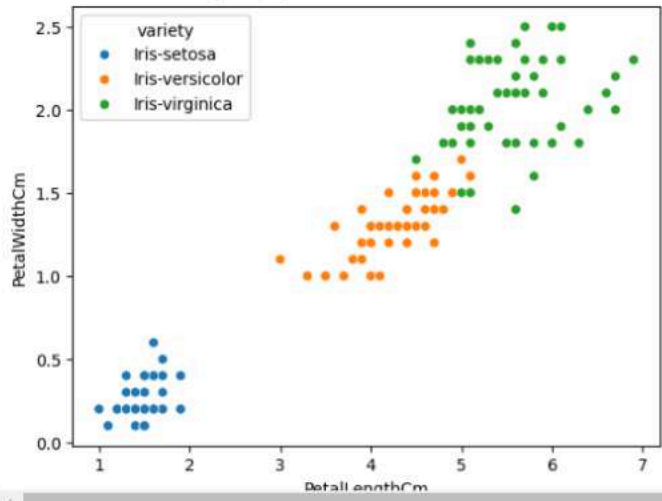
	Iris-setosa	Iris-versicolor	Iris-virginica	Id	SepallengthCm	SepalWidthCm	PetalLengthCm
0	True	False	False	1	5.1	3.5	1.4
1	True	False	False	2	4.9	3.0	1.4
2	True	False	False	3	4.7	3.2	1.3
3	True	False	False	4	4.6	3.1	1.5
4	True	False	False	5	5.0	3.6	1.4

```
sns.scatterplot(x='SepallengthCm',y='SepalWidthCm',hue='variety',data=data,)
```

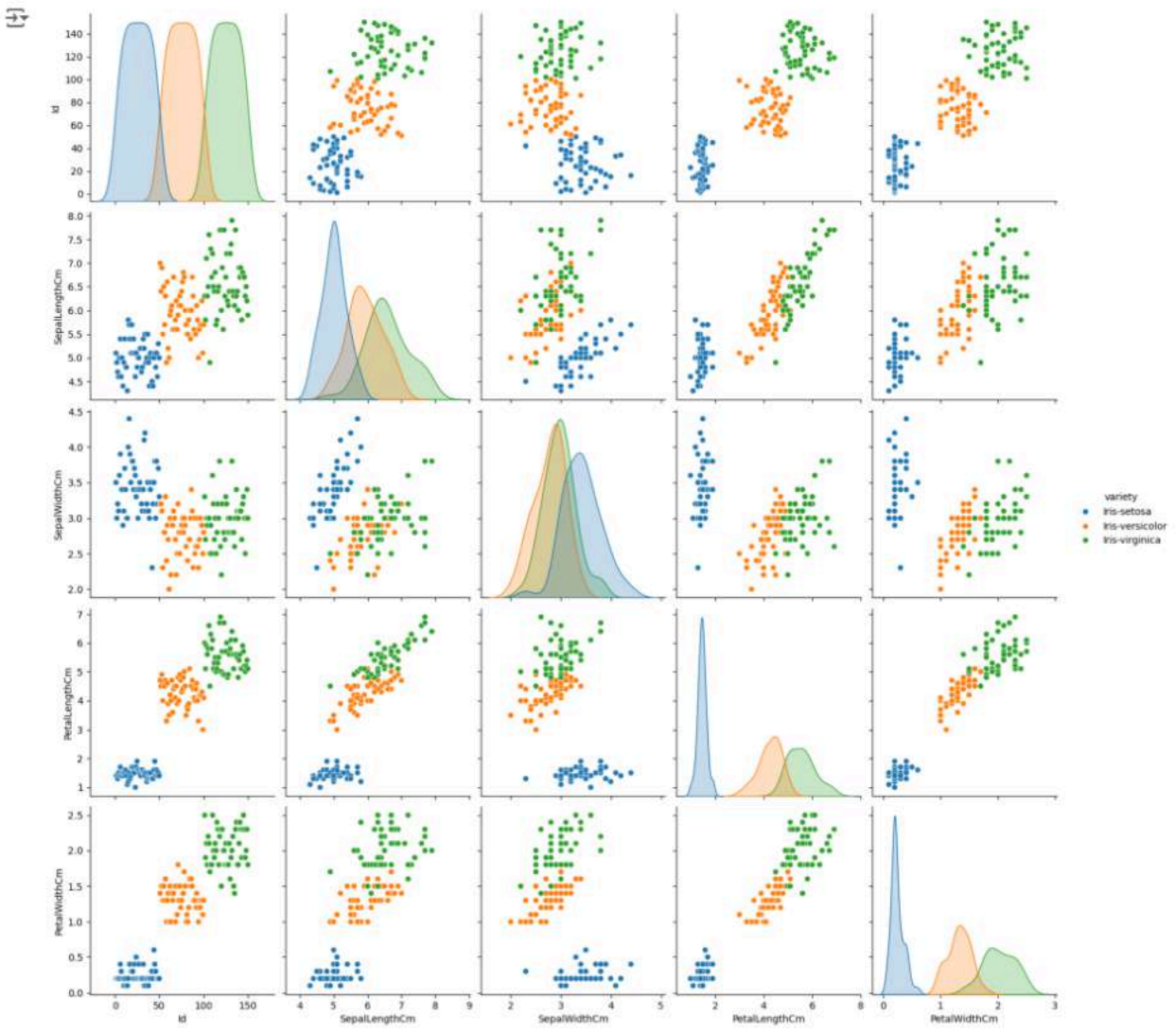


```
sns.scatterplot(x='PetalLengthCm',y='PetalWidthCm',hue='variety',data=data,)
```

```
<Axes: xlabel='PetalLengthCm', ylabel='PetalWidthCm'>
```

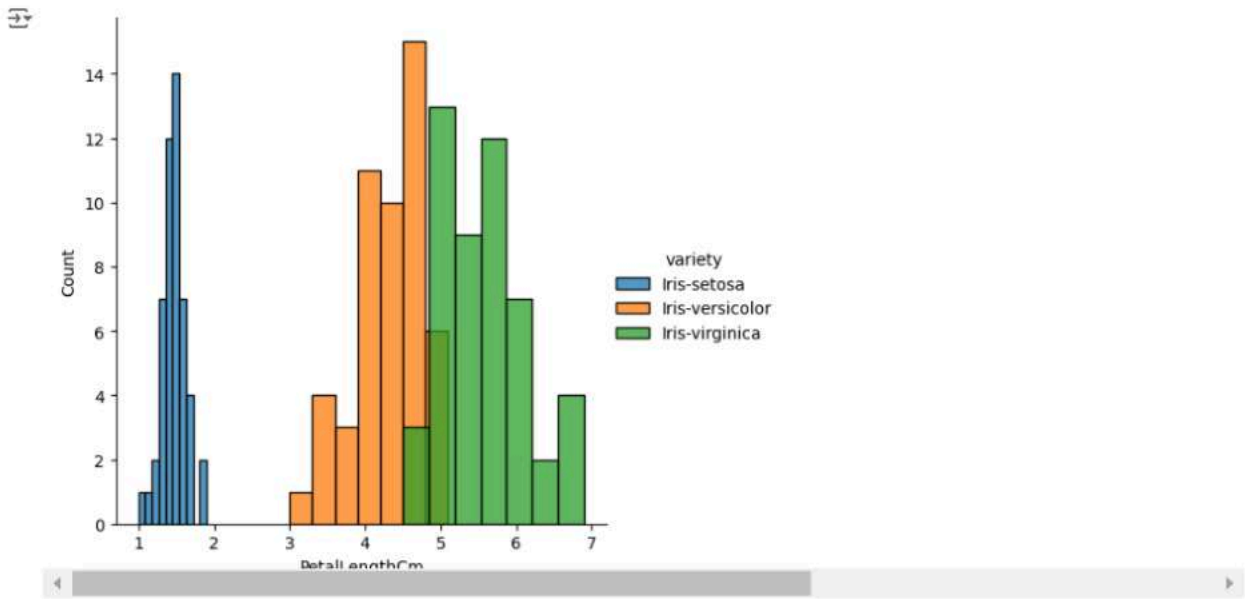


```
sns.pairplot(data,hue='variety',height=3);
```

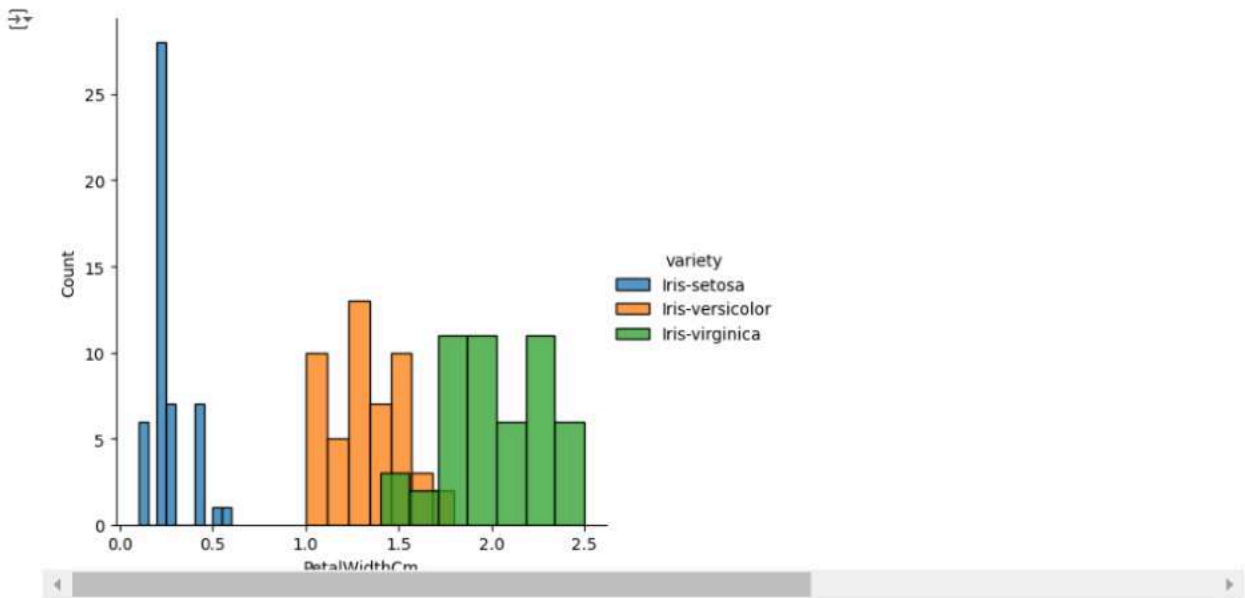


```
plt.show()

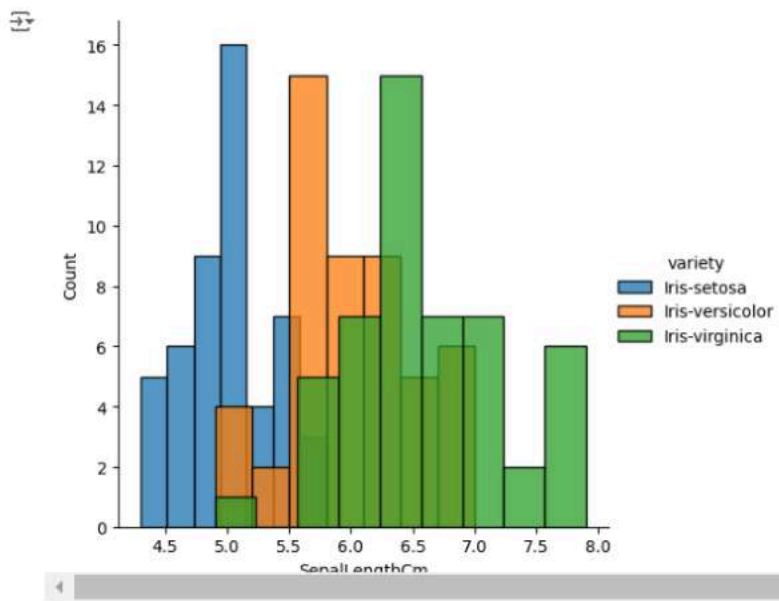
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalLengthCm').add_legend();
plt.show();
```



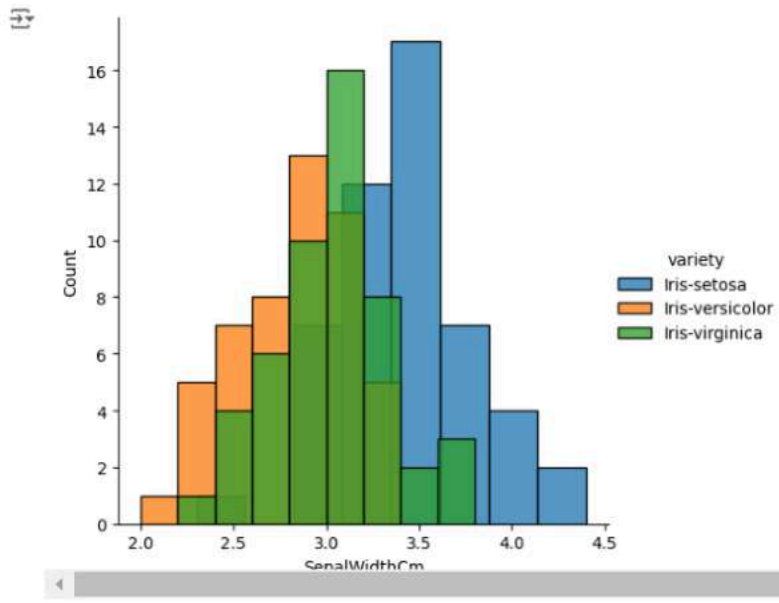
```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalWidthCm').add_legend();
plt.show();
```



```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalLengthCm').add_legend();
plt.show();
```



```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalWidthCm').add_legend();  
plt.show();
```



In []:

```
import numpy as np
array=np.random.randint(1,100,9)
array

array([83, 25, 19, 47, 62, 15, 96, 39, 51])

np.sqrt(array)

array([9.11043358, 5. , 4.35889894, 6.8556546 , 7.87400787,
       3.87298335, 9.79795897, 6.244998 , 7.14142843])

array.ndim

1

new_array=array.reshape(3,3)

new_array

array([[83, 25, 19],
       [47, 62, 15],
       [96, 39, 51]])

new_array.ndim

2

new_array.ravel()

array([83, 25, 19, 47, 62, 15, 96, 39, 51])

newm=new_array.reshape(3,3)

newm

array([[83, 25, 19],
       [47, 62, 15],
       [96, 39, 51]])

newm[2,1:3]

array([39, 51])

newm[1:2,1:3]

array([[62, 15]])

new_array[0:3,0:0]

array([], shape=(3, 0), dtype=int64)

new_array[0:2,0:1]

array([[83],
       [47]])

new_array[0:3,0:1]

array([[83],
       [47],
       [96]])

new_array[1:3]

array([[47, 62, 15],
       [96, 39, 51]])
```

In []:

```
import numpy as np
import pandas as pd
list=[[1, 'Smith',50000],[2, 'Jones',60000]]
```

```
df=pd.DataFrame(list)
df
```

	0	1	2
0	1	Smith	50000
1	2	Jones	60000

```
df.columns=['Empd','Name','Salary']
df
```

	Empd	Name	Salary
0	1	Smith	50000
1	2	Jones	60000

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
#   Column  Non-Null Count  Dtype
---  -
0    Empd      2 non-null      int64
1    Name      2 non-null      object
2    Salary    2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 176.0+ bytes
```

```
df=pd.read_csv("/content/50_Startups.csv")
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
#   Column                Non-Null Count  Dtype
---  -
0    R&D Spend             50 non-null     float64
1    Administration        50 non-null     float64
2    Marketing Spend       50 non-null     float64
3    State                 50 non-null     object
4    Profit                50 non-null     float64
dtypes: float64(4), object(1)
memory usage: 2.1+ KB
```

```
df.head()
```

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	81201.77	366168.42	Florida	166187.04

```
df.tail()
```

	R&D Spend	Administration	Marketing Spend	State	Profit
45	1000.23	124153.04	1903.93	New York	64926.08
46	1315.46	115816.21	297114.46	Florida	49490.75
47	0.00	135426.92	0.00	California	42559.73
48	542.05	51743.15	0.00	New York	35673.41
49	0.00	116082.80	45173.06	California	14681.40

```
import numpy as np
import pandas as pd
df=pd.read_csv("/content/employee.csv")
```

```
df.head()
```

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000

```
df.tail()
```

	emp id	name	salary
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 3 columns):
#   Column  Non-Null Count  Dtype
---  -
0   emp id  7 non-null         int64
1   name    7 non-null         object
2   salary  7 non-null         int64
dtypes: int64(2), object(1)
memory usage: 296.0+ bytes
```

```
df.salary
```

	salary
0	5000
1	6000
2	7000
3	5000
4	8000
5	3000
6	6000

```
type(df.salary)
```

```
pandas.core.series.Series
def __init__(data=None, index=None, dtype: Dtype | None=None, name=None, copy: bool | None=None,
fastpath: bool=False) -> None

One-dimensional ndarray with axis labels (including time series).

Labels need not be unique but must be a hashable type. The object
supports both integer- and label-based indexing and provides a host of
methods for performing operations involving the index. Statistical
```

```
df.salary.mean()
```

```
5714.285714285715
```

```
df.salary.median()
```

```
6000.0
```

```
df.salary.mode()
```

```
salary
0    5000
1    6000
```

```
df.salary.var()
```

```
2571428.5714285714
```

```
df.salary.std()
```

```
1603.5674514745463
```

```
df.describe()
```

```
emp_id  salary
count  7.000000  7.000000
mean    4.000000  5714.285714
std     2.160247  1603.567451
min     1.000000  3000.000000
25%     2.500000  5000.000000
50%     4.000000  6000.000000
75%     5.500000  6500.000000
max     7.000000  8000.000000
```

```
df.describe(include='all')
```

```
emp_id  name  salary
count  7.000000  7  7.000000
unique   NaN  6  NaN
top     NaN  SRI HARSHAVARDHANAN R  NaN
freq     NaN  2  NaN
mean    4.000000  NaN  5714.285714
std     2.160247  NaN  1603.567451
min     1.000000  NaN  3000.000000
25%     2.500000  NaN  5000.000000
50%     4.000000  NaN  6000.000000
75%     5.500000  NaN  6500.000000
max     7.000000  NaN  8000.000000
```

```
empCol=df.columns
```

```
empCol
```

```
Index(['emp_id', 'name ', 'salary'], dtype='object')
```

```
emparray=df.values
```

```
emparray
```

```
array([[1, 'SREE VARSSINI K S', 5000],
       [2, 'SREEMATHI B', 6000],
       [3, 'SREYA G', 7000],
       [4, 'SREYASKARI MULLAPUDI', 5000],
```

In []:

```
[6, 'SRI HARSHAVARDHANAN R', 3000],  
[7, 'SRI HARSHAVARDHANAN R', 6000]], dtype=object)
```

```
employee_DF=pd.DataFrame(emparray,columns=empCol)
```

employee_DF



	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000

Start coding or [generate](#) with AI.

In []:

```
#sample calculation for low range(lr) , upper range (ur),percentile
import numpy as np
array=np.random.randint(1,100,16) # randomly generate 16 numbers between 1 to 100
array
```

```
array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])
```

```
array.mean()
```

```
50.5
```

```
np.percentile(array,25)
```

```
26.0
```

```
np.percentile(array,50)
```

```
56.0
```

```
np.percentile(array,75)
```

```
69.0
```

```
np.percentile(array,100)
```

```
90.0
```

```
#outliers detection
def outDetection(array):
    sorted(array)
    Q1,Q3=np.percentile(array,[25,75])
    IQR=Q3-Q1
    lr=Q1-(1.5*IQR)
    ur=Q3+(1.5*IQR)
    return lr,ur
```

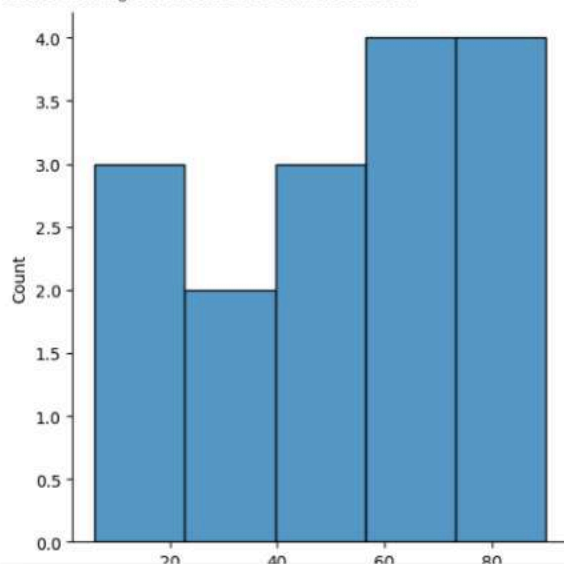
```
lr,ur=outDetection(array)
```

```
lr,ur
```

```
(-38.5, 133.5)
```

```
import seaborn as sns
%matplotlib inline
sns.distplot(array)
```

```
<seaborn.axisgrid.FacetGrid at 0x78f3291c2710>
```



```
sns.distplot(array)
```

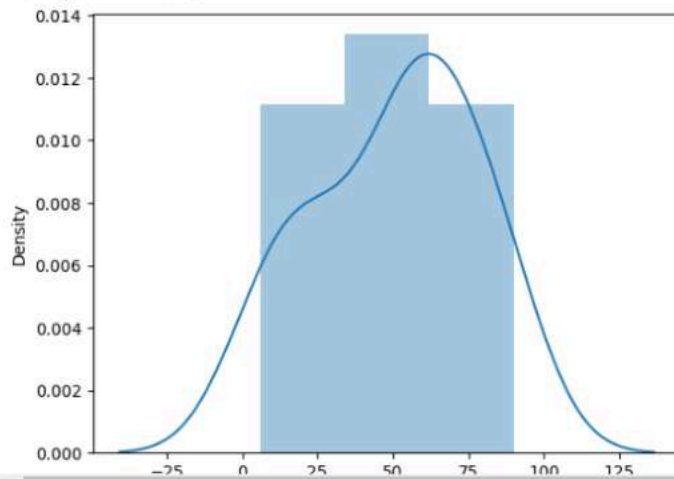
```
>>> <ipython-input-19-d72101983c40>:1: UserWarning:
```

'distplot' is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either 'displot' (a figure-level function with similar flexibility) or 'histplot' (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(array)
<Axes: ylabel='Density'>
```

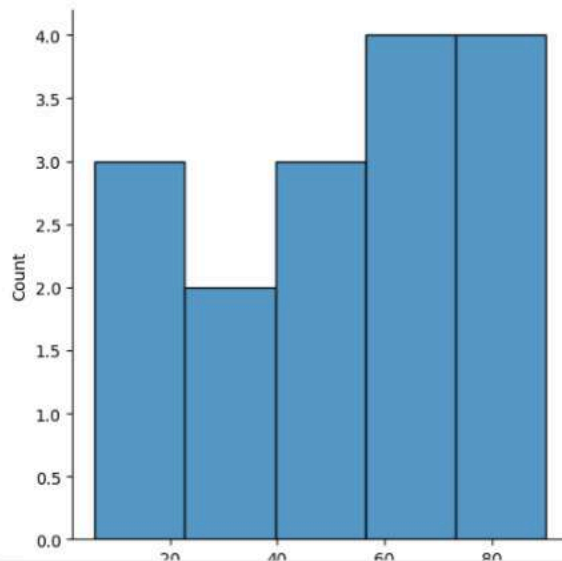


```
new_array=array[(array>lr) & (array<ur)]
new_array
```

```
>>> array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

```
sns.distplot(new_array)
```

```
>>> <seaborn.axisgrid.FacetGrid at 0x78f2e09bb580>
```



```
lr1,ur1=outDetection(new_array)
lr1,ur1
```

```
>>> (-38.5, 133.5)
```

```
final_array=new_array[(new_array>lr1) & (new_array<ur1)]
final_array
```

```
>>> array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

```
sns.distplot(final_array)
```

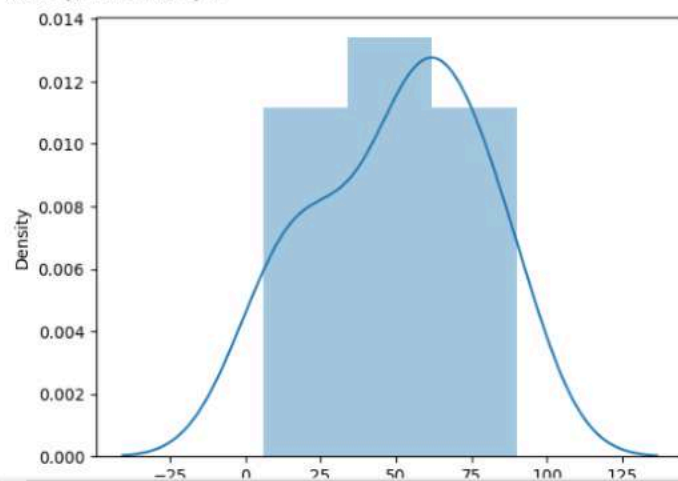
```
<ipython-input-18-7ba96ada5b76>:1: UserWarning:
```

`'distplot'` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `'displot'` (a figure-level function with similar flexibility) or `'histplot'` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see <https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751>

```
sns.distplot(final_array)
<Axes: ylabel='Density'>
```



In []:

Python Code:

Upload Hotel.csv and convert it into dataframe

import numpy as np

import pandas as pd

df=pd.read_csv("Hotel_Dataset.csv")

df

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	Age_Group.1
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30
3	4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+
5	6	35+	3	Ibys	Non-Veg	1909	2	122220	35+
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
7	8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
9	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
10	10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35

#From the dataframe identify the duplicate row(i.e row 9)

The duplicated() method returns a Series with True and False values that describe which rows in the DataFrame are duplicated and not.

df.duplicated()

```
0    False
1    False
2    False
3    False
4    False
5    False
6    False
7    False
8    False
9     True
10   False
dtype: bool
```

The info() method prints information about the DataFrame. The information contains the number of columns, column labels, column data types, memory usage, range index, and the number of cells in each column (non-null values).

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   CustomerID            11 non-null    int64
1   Age_Group              11 non-null    object
2   Rating(1-5)           11 non-null    int64
3   Hotel                  11 non-null    object
4   FoodPreference         11 non-null    object
5   Bill                   11 non-null    int64
6   NoOfPax                11 non-null    int64
7   EstimatedSalary        11 non-null    int64
8   Age_Group.1            11 non-null    object
dtypes: int64(5), object(4)
memory usage: 924.0+ bytes
```

The drop_duplicates() method removes duplicate rows.

```
df.drop_duplicates(inplace=True)
```

```
df
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	Age_Group.1
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30
3	4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+
5	6	35+	3	Ibys	Non-Veg	1909	2	122220	35+
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
7	8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
10	10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35

#While removing duplicate record row index also removed

The len() function to return the length of an object. With a dataframe, the function returns the number of rows.

```
len(df)
```

```
10
```

#Reset the index

```
index=np.array(list(range(0,len(df))))
```

```
df.set_index(index,inplace=True)
```

```
index
```

```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

```
df
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	Age_Group.1
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30
3	4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+
5	6	35+	3	lbys	Non-Veg	1909	2	122220	35+
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
7	8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
9	10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35

Axis refers to the dimensions of a DataFrame (index and columns) or Series (index only) Use axis=0 to apply functions row-wise along the index. Use axis=1 to apply functions column-wise across columns.

```
df.drop(['Age_Group.1'],axis=1,inplace=True)
```

```
df
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1	20-25	4	Ibis	veg	1300	2	40000
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000
2	3	25-30	6	RedFox	Veg	1322	2	30000
3	4	20-25	-1	LemonTree	Veg	1234	2	120000
4	5	35+	3	Ibis	Vegetarian	989	2	45000
5	6	35+	3	lbys	Non-Veg	1909	2	122220
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122
7	8	20-25	7	LemonTree	Veg	2999	-10	345673
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999
9	10	30-35	5	RedFox	non-Veg	-6755	4	87777

The function .loc is typically used for label indexing and can access multiple columns.

```
df.CustomerID.loc[df.CustomerID<0]=np.nan
```

```
df.Bill.loc[df.Bill<0]=np.nan
```

```
df.EstimatedSalary.loc[df.EstimatedSalary<0]=np.nan
```

```
df
```

```
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2580639570.py:1: SettingWithCopyWarning:
```

A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df.CustomerID.loc[df.CustomerID<0]=np.nan
```

```
df.Bill.loc[df.Bill<0]=np.nan
```

```
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2580639570.py:2: SettingWithCopyWarning:
```

A value is trying to be set on a copy of a slice from a DataFrame

```
df.EstimatedSalary.loc[df.EstimatedSalary<0]=np.nan
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4.0	Ibis	veg	1300.0	2	40000.0
1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0	3	59000.0
2	3.0	25-30	NaN	RedFox	Veg	1322.0	2	30000.0
3	4.0	20-25	NaN	LemonTree	Veg	1234.0	2	120000.0
4	5.0	35+	3.0	Ibis	Vegetarian	989.0	2	45000.0
5	6.0	35+	3.0	Ibys	Non-Veg	1909.0	2	122220.0
6	7.0	35+	4.0	RedFox	Vegetarian	1000.0	-1	21122.0
7	8.0	20-25	NaN	LemonTree	Veg	2999.0	-10	345673.0
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	3	NaN
9	10.0	30-35	5.0	RedFox	non-Veg	NaN	4	87777.0

```
df['NoOfPax'].loc[(df['NoOfPax']<1) | (df['NoOfPax']>20)]=np.nan
```

```
df
```

```
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2129877948.py:1: SettingWithCopyWarning:
```

A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy

```
df['NoOfPax'].loc[(df['NoOfPax']<1) | (df['NoOfPax']>20)]=np.nan
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4.0	Ibis	veg	1300.0	2.0	40000.0
1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0	3.0	59000.0
2	3.0	25-30	NaN	RedFox	Veg	1322.0	2.0	30000.0
3	4.0	20-25	NaN	LemonTree	Veg	1234.0	2.0	120000.0
4	5.0	35+	3.0	Ibis	Vegetarian	989.0	2.0	45000.0
5	6.0	35+	3.0	Ibys	Non-Veg	1909.0	2.0	122220.0
6	7.0	35+	4.0	RedFox	Vegetarian	1000.0	NaN	21122.0
7	8.0	20-25	NaN	LemonTree	Veg	2999.0	NaN	345673.0
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	3.0	NaN
9	10.0	30-35	5.0	RedFox	non-Veg	NaN	4.0	87777.0

df.Age_Group.unique()

```
array(['20-25', '30-35', '25-30', '35+'], dtype=object)
```

df.Hotel.unique()

```
array(['Ibis', 'LemonTree', 'RedFox', 'Ibys'], dtype=object)
```

Using the inplace=True keyword in a pandas method changes the default behaviour such that the operation on the dataframe doesn't return anything, it instead 'modifies the underlying data

df.Hotel.replace(['Ibys'],'Ibis',inplace=True)

df.FoodPreference.unique

```
<bound method Series.unique of 0          veg
```

```
1      Non-Veg
```

```
2          Veg
```

```
3          Veg
```

```
4  Vegetarian
```

```
5      Non-Veg
```

```
6  Vegetarian
```

```
7          Veg
```

```
8      Non-Veg
```

```
9      non-Veg
```

```
Name: FoodPreference, dtype: object>
```

df.FoodPreference.replace(['Vegetarian','veg'],'Veg',inplace=True)

df.FoodPreference.replace(['non-Veg'],'Non-Veg',inplace=True)

Fillna is a Pandas function to fill the NA/NaN values with the specified method.

If column or feature is numerical continuous data then replace the missing(NaN) value by taking mean value.

If column or feature is numerical discrete data then replace the missing(NaN) value by taking median value.

If column or feature is non-numerical i.e Categorical data then replace the missing(NaN) value by taking mode value.

```
df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()),inplace=True)
```

```
df.NoOfPax.fillna(round(df.NoOfPax.median()),inplace=True)
```

```
df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()), inplace=True)
```

```
df.Bill.fillna(round(df.Bill.mean()),inplace=True)
```

```
df
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Estimated Salary
0	1.0	20-25	4.0	Ibis	Veg	1300.0	2.0	40000.0
1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0	3.0	59000.0
2	3.0	25-30	4.0	RedFox	Veg	1322.0	2.0	30000.0
3	4.0	20-25	4.0	LemonTree	Veg	1234.0	2.0	120000.0
4	5.0	35+	3.0	Ibis	Veg	989.0	2.0	45000.0
5	6.0	35+	3.0	Ibis	Non-Veg	1909.0	2.0	122220.0
6	7.0	35+	4.0	RedFox	Veg	1000.0	2.0	21122.0
7	8.0	20-25	4.0	LemonTree	Veg	2999.0	2.0	345673.0
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	3.0	96755.0
9	10.0	30-35	5.0	RedFox	Non-Veg	1801.0	4.0	87777.0

In []:

```
import numpy as np
import pandas as pd
df=pd.read_csv('/content/pre-process_datasample.csv')
```

df



	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	NaN	50.0	83000.0	No
9	France	37.0	67000.0	Yes

Next steps:

[Generate code with df](#)

 [View recommended plots](#)

[New interactive sheet](#)

df.head()



	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes

Next steps:

[Generate code with df](#)

 [View recommended plots](#)

[New interactive sheet](#)

```
df.Country.fillna(df.Country.mode()[0],inplace=True)
features=df.iloc[:,:-1].values

<ipython-input-5-20665a0bbaa1>:1: FutureWarning: A value is trying to be set on a copy of a DataFrame c
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate of

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)'

df.Country.fillna(df.Country.mode()[0],inplace=True)
```

label=df.iloc[:, -1].values

```

from sklearn.impute import SimpleImputer

age=SimpleImputer(strategy="mean",missing_values=np.nan)

Salary=SimpleImputer(strategy="mean",missing_values=np.nan)

age.fit(features[:,[1]])

```



```

SimpleImputer ⓘ ?
SimpleImputer()

```

```
Salary.fit(features[:,[2]])
```



```

SimpleImputer ⓘ ?
SimpleImputer()

```

```
SimpleImputer()
```



```

SimpleImputer ⓘ ?
SimpleImputer()

```

```
features[:,[1]]=age.transform(features[:,[1]])
```

```
features[:,[2]]=Salary.transform(features[:,[2]])
```

```
features
```



```

array([[ 'France', 44.0, 72000.0],
       [ 'Spain', 27.0, 48000.0],
       [ 'Germany', 30.0, 54000.0],
       [ 'Spain', 38.0, 61000.0],
       [ 'Germany', 40.0, 63777.77777777778],
       [ 'France', 35.0, 58000.0],
       [ 'Spain', 38.77777777777778, 52000.0],
       [ 'France', 48.0, 79000.0],
       [ 'France', 50.0, 83000.0],
       [ 'France', 37.0, 67000.0]], dtype=object)

```

```
from sklearn.preprocessing import OneHotEncoder
```

```
oh = OneHotEncoder(sparse_output=False)
```

```
Country=oh.fit_transform(features[:,[0]])
```

```
Country
```



```

array([[1., 0., 0.],
       [0., 0., 1.],
       [0., 1., 0.],
       [0., 0., 1.],
       [0., 1., 0.],
       [1., 0., 0.],
       [0., 0., 1.],
       [1., 0., 0.]])

```

```
[1., 0., 0.],
[1., 0., 0.]])
```

```
final_set=np.concatenate((Country,features[:,[1,2]]),axis=1)
```

```
final_set
```

```
array([[1.0, 0.0, 0.0, 44.0, 72000.0],
       [0.0, 0.0, 1.0, 27.0, 48000.0],
       [0.0, 1.0, 0.0, 30.0, 54000.0],
       [0.0, 0.0, 1.0, 38.0, 61000.0],
       [0.0, 1.0, 0.0, 40.0, 63777.77777777778],
       [1.0, 0.0, 0.0, 35.0, 58000.0],
       [0.0, 0.0, 1.0, 38.77777777777778, 52000.0],
       [1.0, 0.0, 0.0, 48.0, 79000.0],
       [1.0, 0.0, 0.0, 50.0, 83000.0],
       [1.0, 0.0, 0.0, 37.0, 67000.0]], dtype=object)
```

```
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
sc.fit(final_set)
feat_standard_scaler=sc.transform(final_set)
```

```
feat_standard_scaler
```

```
array([[ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
         7.58874362e-01,  7.49473254e-01],
       [-1.00000000e+00, -5.00000000e-01,  1.52752523e+00,
        -1.71150388e+00, -1.43817841e+00],
       [-1.00000000e+00,  2.00000000e+00, -6.54653671e-01,
        -1.27555478e+00, -8.91265492e-01],
       [-1.00000000e+00, -5.00000000e-01,  1.52752523e+00,
        -1.13023841e-01, -2.53200424e-01],
       [-1.00000000e+00,  2.00000000e+00, -6.54653671e-01,
         1.77608893e-01,  6.63219199e-16],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
        -5.48972942e-01, -5.26656882e-01],
       [-1.00000000e+00, -5.00000000e-01,  1.52752523e+00,
         0.00000000e+00, -1.07356980e+00],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
         1.34013983e+00,  1.38753832e+00],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
         1.63077256e+00,  1.75214693e+00],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
        -2.58340208e-01,  2.93712492e-01]])
```

```
from sklearn.preprocessing import MinMaxScaler
mms=MinMaxScaler(feature_range=(0,1))
mms.fit(final_set)
feat_minmax_scaler=mms.transform(final_set)
feat_minmax_scaler
```

```
array([[1., 0., 0., 0.73913043, 0.68571429],
       [0., 0., 1., 0., 0.],
       [0., 1., 0., 0.13043478, 0.17142857],
       [0., 0., 1., 0.47826087, 0.37142857],
       [0., 1., 0., 0.56521739, 0.45079365],
       [1., 0., 0., 0.34782609, 0.28571429],
       [0., 0., 1., 0.51207729, 0.11428571],
       [1., 0., 0., 0.91304348, 0.88571429],
       [1., 0., 0., 1., 1.],
       [1., 0., 0., 0.43478261, 0.54285714]])
```

In []:

```
import numpy as np
import pandas as pd
df=pd.read_csv("/content/pre-process_datasample.csv")
df
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	NaN	50.0	83000.0	No
9	France	37.0	67000.0	Yes

Double-click (or enter) to edit

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Country     9 non-null      object
1   Age         9 non-null      float64
2   Salary      9 non-null      float64
3   Purchased   10 non-null     object
dtypes: float64(2), object(2)
memory usage: 448.0+ bytes
```

```
df.Country.mode()
```

Country
0 France

```
df.Country.mode()[0]
```

```
type(df.Country.mode())
```

```
pandas.core.series.Series
def __init__(data=None, index=None, dtype: Dtype | None=None, name=None, copy: bool | None=None,
fastpath: bool=False) -> None
index is None, the resulting series is indexed with the index values.
dtype : str, numpy.dtype, or ExtensionDtype, optional
Data type for the output Series. If not specified, this will be
inferred from 'data'.
See the :ref:'user guide <basics.dtypes>' for more usages.
name : Hashable, default None
The name to give to the Series.
```

```
df.Country.fillna(df.Country.mode()[0],inplace=True)
```

```
df.Age.fillna(df.Age.median(),inplace=True)
```

```
df.Salary.fillna(round(df.Salary.mean()),inplace=True)
```

```
df
```

```
In []:
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	63778.0	Yes
5	France	35.0	58000.0	Yes
6	Spain	38.0	52000.0	No
7	France	48.0	79000.0	Yes
8	France	50.0	83000.0	No
9	France	37.0	67000.0	Yes

```
pd.get_dummies(df.Country)
```

	France	Germany	Spain
0	True	False	False
1	False	False	True
2	False	True	False
3	False	False	True
4	False	True	False
5	True	False	False
6	False	False	True
7	True	False	False
8	True	False	False
9	True	False	False

```
updated_dataset=pd.concat([pd.get_dummies(df.Country),df.iloc[:,[1,2,3]]],axis=1)
```

```
updated_dataset
```

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	No
1	False	False	True	27.0	48000.0	Yes
2	False	True	False	30.0	54000.0	No
3	False	False	True	38.0	61000.0	No
4	False	True	False	40.0	63778.0	Yes
5	True	False	False	35.0	58000.0	Yes
6	False	False	True	38.0	52000.0	No
7	True	False	False	48.0	79000.0	Yes
8	True	False	False	50.0	83000.0	No
9	True	False	False	37.0	67000.0	Yes

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Country     10 non-null    object
1   Age         10 non-null    float64
2   Salary      10 non-null    float64
3   Purchased   10 non-null    object
dtypes: float64(2), object(2)
memory usage: 448.0+ bytes
```

```
updated_dataset.Purchased.replace(['No','Yes'],[0,1],inplace=True)
```

updated_dataset



	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	0
1	False	False	True	27.0	48000.0	1
2	False	True	False	30.0	54000.0	0
3	False	False	True	38.0	61000.0	0
4	False	True	False	40.0	63778.0	1
5	True	False	False	35.0	58000.0	1
6	False	False	True	38.0	52000.0	0
7	True	False	False	48.0	79000.0	1
8	True	False	False	50.0	83000.0	0
9	True	False	False	37.0	67000.0	1

Start coding or [generate](#) with AI.

In []:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline
```

```
tips=sns.load_dataset('tips')
```

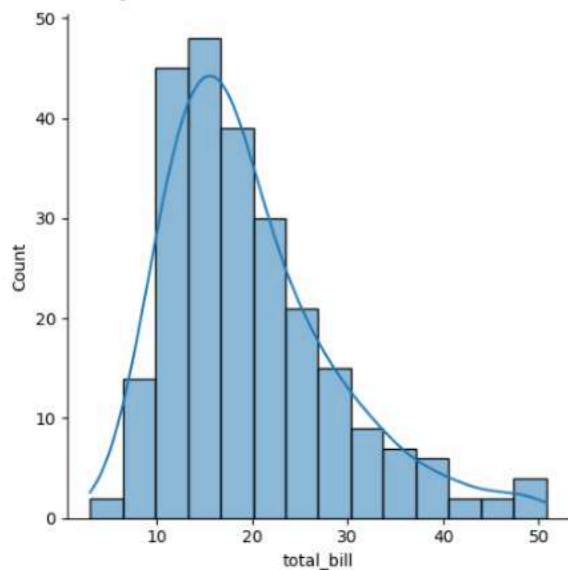
```
tips.head()
```

```
total_bill  tip    sex  smoker  day  time  size
0      16.99   1.01 Female     No  Sun  Dinner     2
1      10.34   1.66   Male     No  Sun  Dinner     3
2      21.01   3.50   Male     No  Sun  Dinner     3
3      23.68   3.31   Male     No  Sun  Dinner     2
4      24.59   3.61 Female     No  Sun  Dinner     4
```

[+ Code](#)[+ Text](#)

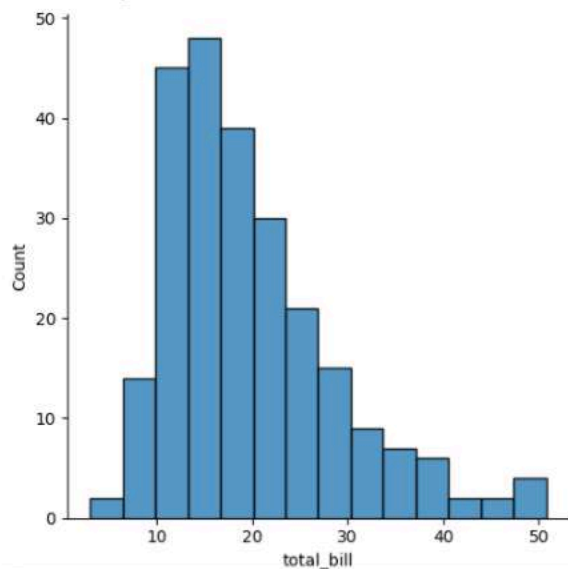
```
sns.displot(tips.total_bill,kde=True)
```

```
<seaborn.axisgrid.FacetGrid at 0x79bb4c7ea680>
```



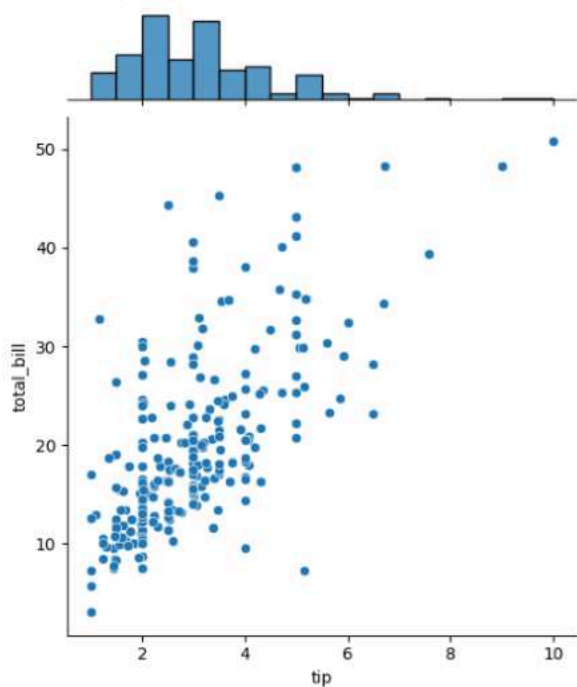
```
sns.displot(tips.total_bill,kde=False)
```

```
<seaborn.axisgrid.FacetGrid at 0x79bb0b0af580>
```



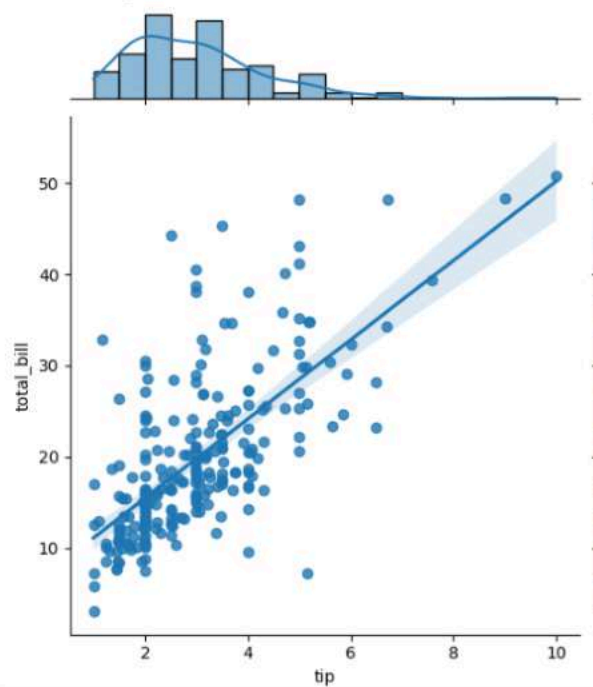
```
sns.jointplot(x=tips.tip,y=tips.total_bill)
```

```
<seaborn.axisgrid.JointGrid at 0x79bb08fc96c0>
```



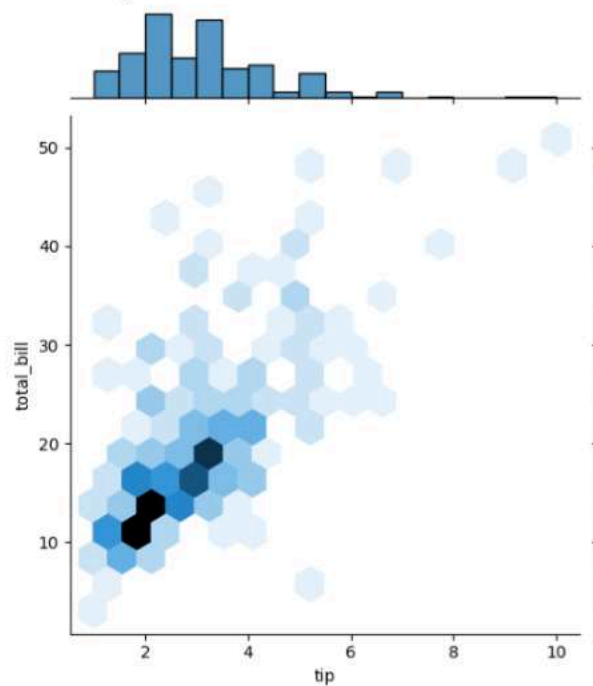
```
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="reg")
```

```
<seaborn.axisgrid.JointGrid at 0x79bb08fc9cf0>
```



```
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="hex")
```

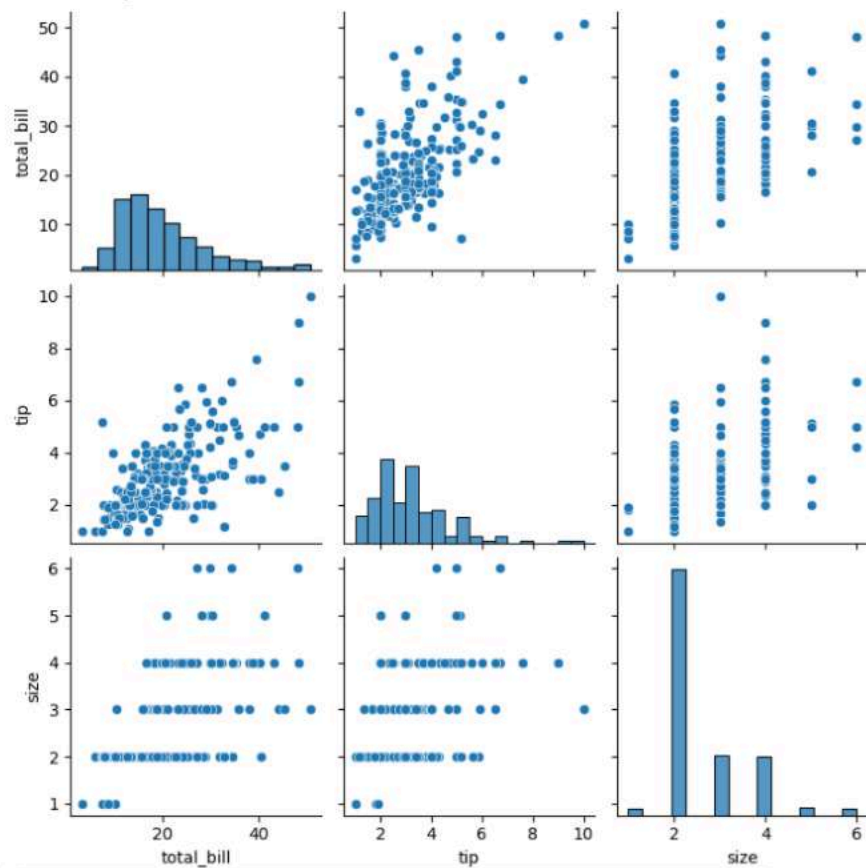
```
<seaborn.axisgrid.JointGrid at 0x79bb088f4730>
```



```
sns.pairplot(tips)
```

In []:

```
<seaborn.axisgrid.PairGrid at 0x79bb06fc3d30>
```



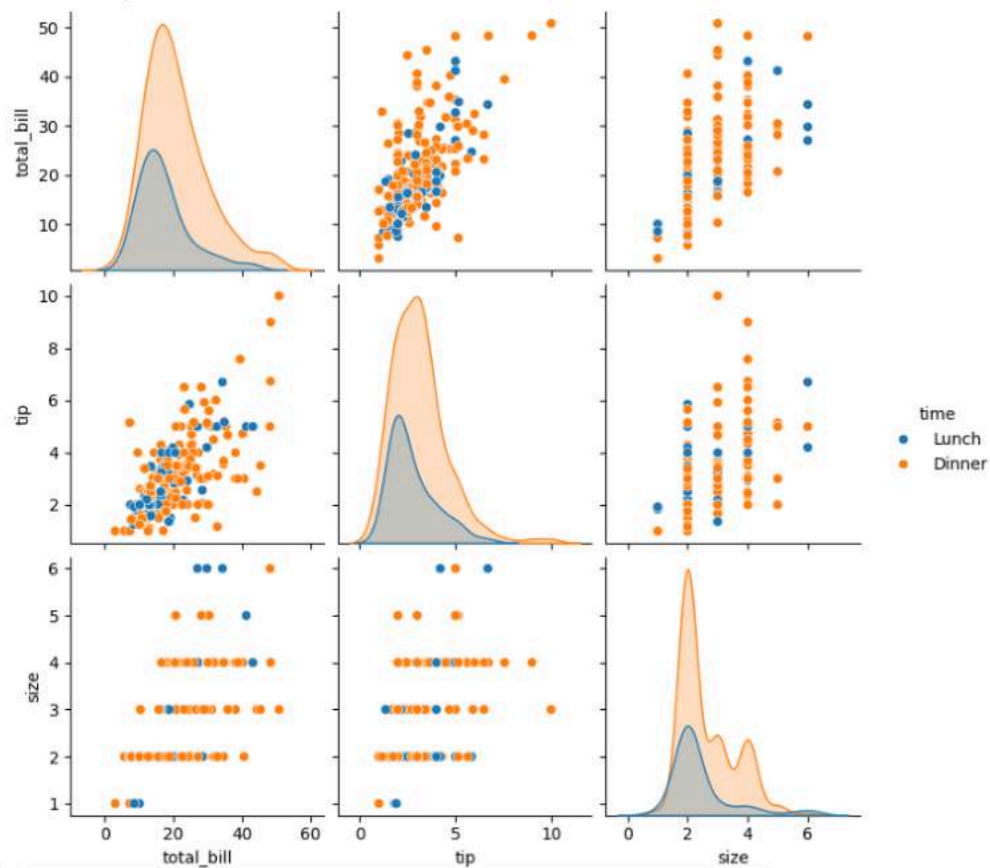
```
tips.time.value_counts()
```

```
count
time
Dinner    176
Lunch      68
```

```
dtype: int64
```

```
sns.pairplot(tips,hue='time')
```

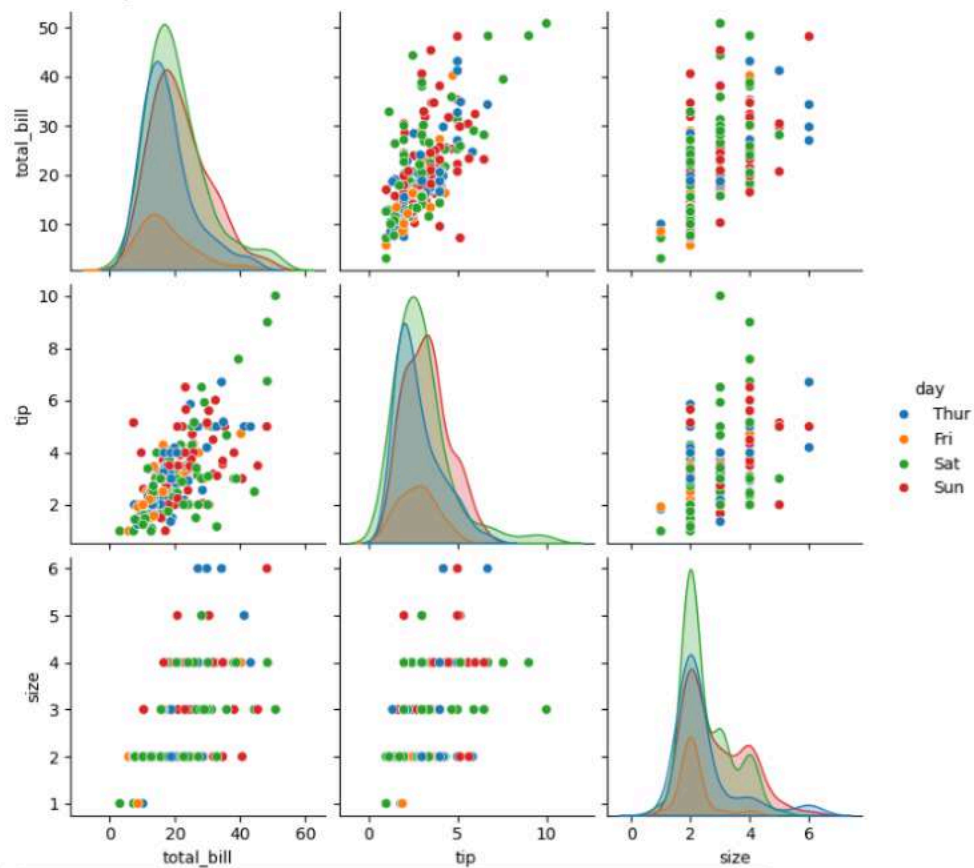
```
<seaborn.axisgrid.PairGrid at 0x79bb088f4670>
```



```
sns.pairplot(tips, hue='day')
```

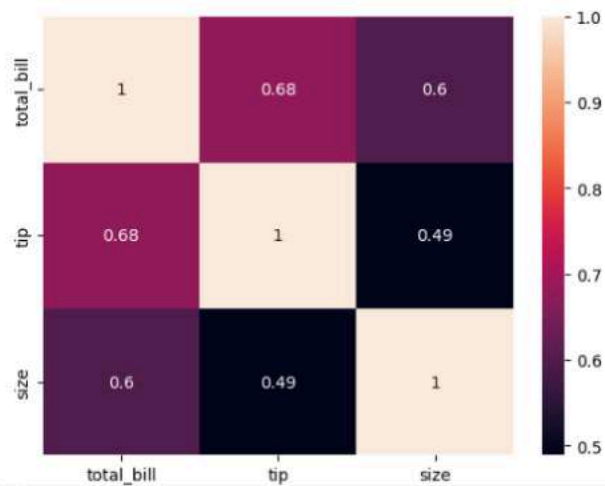
```
In [ ]:
```

```
<seaborn.axisgrid.PairGrid at 0x79bb08f1f6a0>
```



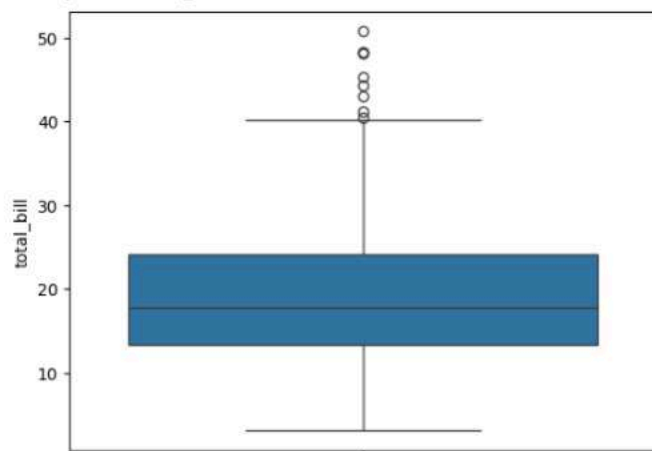
```
sns.heatmap(tips.corr(numeric_only=True),annot=True)
```

```
<Axes: >
```



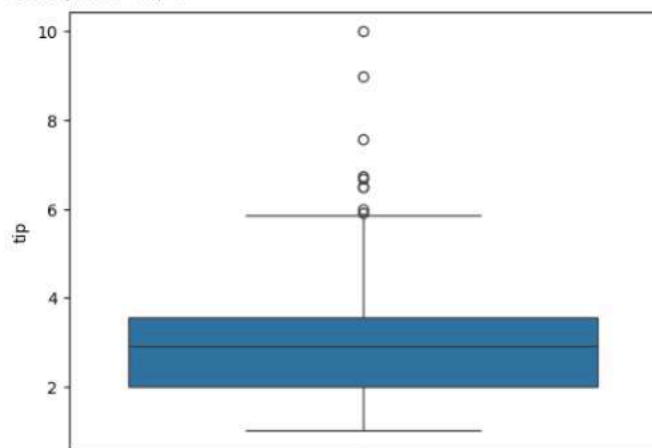
```
sns.boxplot(tips.total_bill)
```

```
sns.boxplot(tips.total_bill)
```



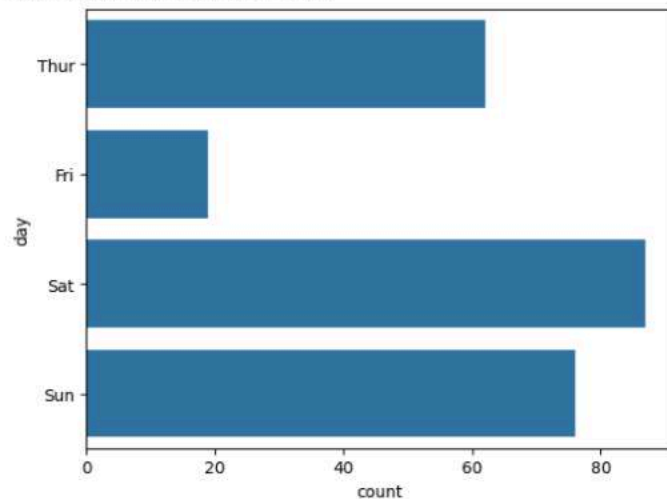
```
sns.boxplot(tips.tip)
```

```
<Axes: ylabel='tip'>
```



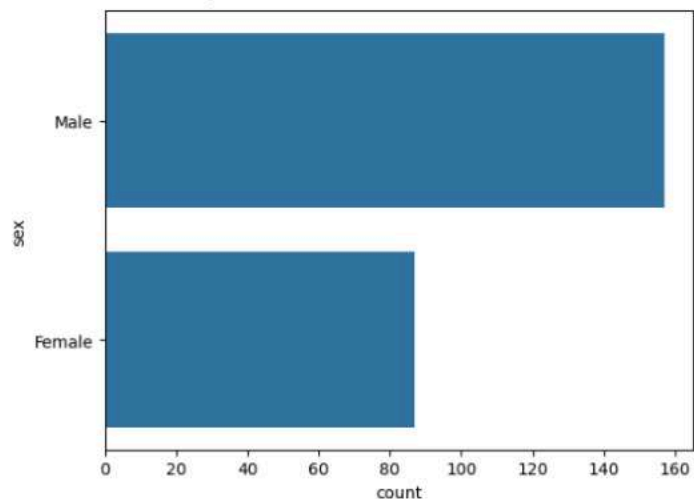
```
sns.countplot(tips.day)
```

```
<Axes: xlabel='count', ylabel='day'>
```



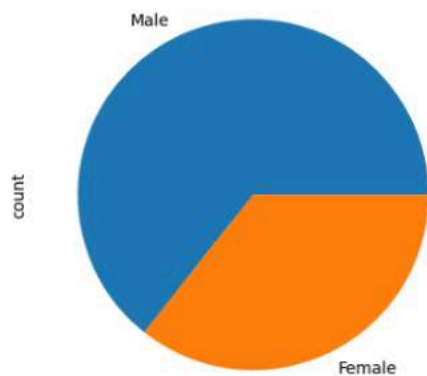
```
sns.countplot(tips.sex)
```

```
<Axes: xlabel='count', ylabel='sex'>
```



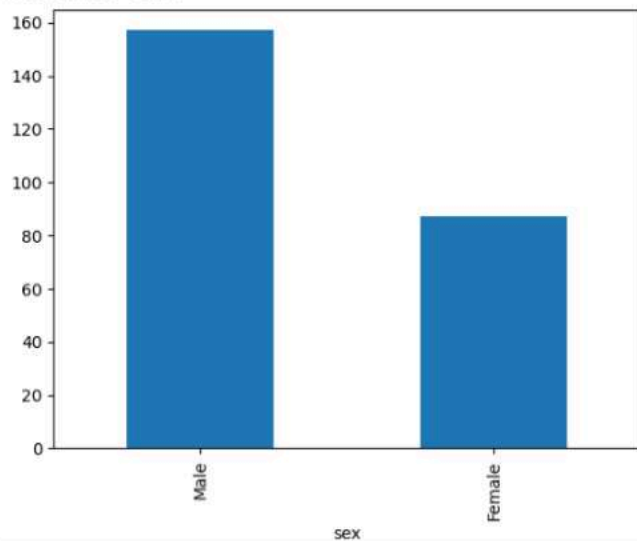
```
tips.sex.value_counts().plot(kind='pie')
```

```
<Axes: ylabel='count'>
```



```
tips.sex.value_counts().plot(kind='bar')
```

```
<Axes: xlabel='sex'>
```



```
sns.countplot(tips[tips.time=='Dinner']['day'])
```

<Axes: xlabel='count', ylabel='day'>



In []:

```
In [ ]: import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df
```

```
In [19]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   YearsExperience  30 non-null    float64
1   Salary          30 non-null    int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

```
In [3]: df.dropna(inplace=True)
```

```
In [4]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   YearsExperience  30 non-null    float64
1   Salary          30 non-null    int64
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

```
In [5]: df.describe()
```

```
Out[5]:
```

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

```
In [6]: features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
```

```
In [7]: from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_st
```

```
In [20]: from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
```

```
Out[20]:
```

LinearRegression

LinearRegression()

```
In [21]: model.score(x_train,y_train)
```

```
Out[21]: 0.9603182547438908
```

```
In [23]: model.score(x_test,y_test)
```

```
Out[23]: 0.9184170849214232
```

```
In [24]: model.coef_
```

```
Out[24]: array([[9281.30847068]])
```

```
In [25]: model.intercept_
```

```
Out[25]: array([27166.73682891])
```

```
In [26]: import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
```

```
In [27]: model=pickle.load(open('SalaryPred.model','rb'))
```

```
In [28]: yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)
```

```
Enter Years of Experience: 44
```

```
In [ ]:
```

```
In [29]: print("Estimated Salary for {} years of experience is {}: " .format(yr_of_exp,Salary))
```

```
Estimated Salary for 44.0 years of experience is [[435544.30953887]]:
```

```
In [ ]:
```

In []:

```
In [1]: import numpy as np
import pandas as pd
```

```
In [2]: df=pd.read_csv('Iris.csv')
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   sepal.length    150 non-null    float64
1   sepal.width     150 non-null    float64
2   petal.length    150 non-null    float64
3   petal.width     150 non-null    float64
4   variety         150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

```
In [3]: df.variety.value_counts()
```

```
Out[3]: Setosa      50
Versicolor  50
Virginica    50
Name: variety, dtype: int64
```

```
In [4]: df.head()
```

```
Out[4]:
```

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

```
In [5]: features=df.iloc[:, :-1].values
label=df.iloc[:, 4].values
```

```
In [6]: from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
```

```
In [8]: xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2,random_state=0)
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)
```

```
Out[8]: KNeighborsClassifier()
```

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.

On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

```
In [9]: print(model_KNN.score(xtrain,ytrain))
        print(model_KNN.score(xtest,ytest))
```

```
0.9583333333333334
1.0
```

```
In [10]: from sklearn.metrics import confusion_matrix
         confusion_matrix(label,model_KNN.predict(features))
```

```
Out[10]: array([[50,  0,  0],
                [ 0, 47,  3],
                [ 0,  2, 48]], dtype=int64)
```

```
In [11]: from sklearn.metrics import classification_report
         print(classification_report(label,model_KNN.predict(features)))
```

	precision	recall	f1-score	support
Setosa	1.00	1.00	1.00	50
Versicolor	0.96	0.94	0.95	50
Virginica	0.94	0.96	0.95	50
accuracy			0.97	150
macro avg	0.97	0.97	0.97	150
weighted avg	0.97	0.97	0.97	150

```
In [ ]:
```

In []:

```
In [1]: import numpy as np
import pandas as pd
df=pd.read_csv('Social_Network_Ads.csv')
df
```

Out[1]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0
...	...	...	...	...	...
395	15691863	Female	46	41000	1
396	15706071	Male	51	23000	1
397	15654296	Female	50	20000	1
398	15755018	Male	36	33000	0
399	15594041	Female	49	36000	1

400 rows × 5 columns

```
In [2]: df.head()
```

Out[2]:

	User ID	Gender	Age	EstimatedSalary	Purchased
0	15624510	Male	19	19000	0
1	15810944	Male	35	20000	0
2	15668575	Female	26	43000	0
3	15603246	Female	27	57000	0
4	15804002	Male	19	76000	0

```
In [4]: features=df.iloc[:,[2,3]].values
label=df.iloc[:,4].values
features
```

```
Out[4]: array([[ 19, 19000],
 [ 35, 20000],
 [ 26, 43000],
 [ 27, 57000],
 [ 19, 76000],
 [ 27, 58000],
 [ 27, 84000],
 [ 32, 150000],
 [ 25, 33000],
 [ 35, 65000],
 [ 26, 80000],
 [ 26, 52000],
 [ 20, 86000],
 [ 32, 18000],
 [ 18, 82000],
 [ 29, 80000],
 [ 47, 25000],
 [ 45, 26000],
 [ 46, 28000],
 [ 16, 20000]])
```

```
In [5]: label
```

```
Out[5]: array([0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 1, 1, 1, 1, 1,
               1, 1, 1, 1, 1, 1, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
               0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,
               0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0,
               0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,
               0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
               0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,
               0, 0, 0, 0, 0, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0,
               0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
               0, 0, 0, 0, 1, 0, 1, 0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 0, 0, 0, 1,
               0, 1, 1, 1, 0, 0, 1, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 0, 0, 1, 1, 0,
               1, 0, 1, 0, 1, 0, 1, 0, 1, 0, 0, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1,
               1, 1, 0, 0, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 0, 1,
               0, 1, 0, 1, 1, 1, 1, 0, 0, 0, 1, 1, 0, 1, 1, 0, 1, 1, 1, 0, 0, 1,
               1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1, 1, 0, 1, 1, 1, 1,
               0, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 0,
               1, 1, 0, 1], dtype=int64)
```

```
In [6]: from sklearn.model_selection import train_test_split
        from sklearn.linear_model import LogisticRegression
```

```
In [7]: for i in range(1,401):
        x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,
        model=LogisticRegression()
        model.fit(x_train,y_train)
        train_score=model.score(x_train,y_train)
        test_score=model.score(x_test,y_test)
        if test_score>train_score:
            print("Test {} Train{} Random State {}".format(test_score,train_score,i))
```

```
Test 0.6875 Train0.63125 Random State 3
Test 0.7375 Train0.61875 Random State 4
Test 0.6625 Train0.6375 Random State 5
Test 0.65 Train0.640625 Random State 6
Test 0.675 Train0.634375 Random State 7
Test 0.675 Train0.634375 Random State 8
Test 0.65 Train0.640625 Random State 10
Test 0.6625 Train0.6375 Random State 11
Test 0.7125 Train0.625 Random State 13
Test 0.675 Train0.634375 Random State 16
Test 0.7 Train0.628125 Random State 17
Test 0.7 Train0.628125 Random State 21
Test 0.65 Train0.640625 Random State 24
Test 0.6625 Train0.6375 Random State 25
Test 0.75 Train0.615625 Random State 26
Test 0.675 Train0.634375 Random State 27
Test 0.7 Train0.628125 Random State 28
Test 0.6875 Train0.63125 Random State 29
Test 0.6875 Train0.63125 Random State 31
```

```
In [8]: x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,
        finalModel=LogisticRegression()
        finalModel.fit(x_train,y_train)
```

Out[8]: LogisticRegression()

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.

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```
In [9]: print(finalModel.score(x_train,y_train))
        print(finalModel.score(x_test,y_test))
```

```
0.834375
0.9125
```

```
In [10]: from sklearn.metrics import classification_report
         print(classification_report(label,finalModel.predict(features)))
```

	precision	recall	f1-score	support
0	0.85	0.93	0.89	257
1	0.84	0.71	0.77	143
accuracy			0.85	400
macro avg	0.85	0.82	0.83	400
weighted avg	0.85	0.85	0.85	400

In []:

In []:

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
In [2]: df=pd.read_csv('Mall_Customers.csv')
```

```
In [3]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   CustomerID                           200 non-null    int64
1   Gender                               200 non-null    object
2   Age                                   200 non-null    int64
3   Annual Income (k$)                   200 non-null    int64
4   Spending Score (1-100)               200 non-null    int64
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

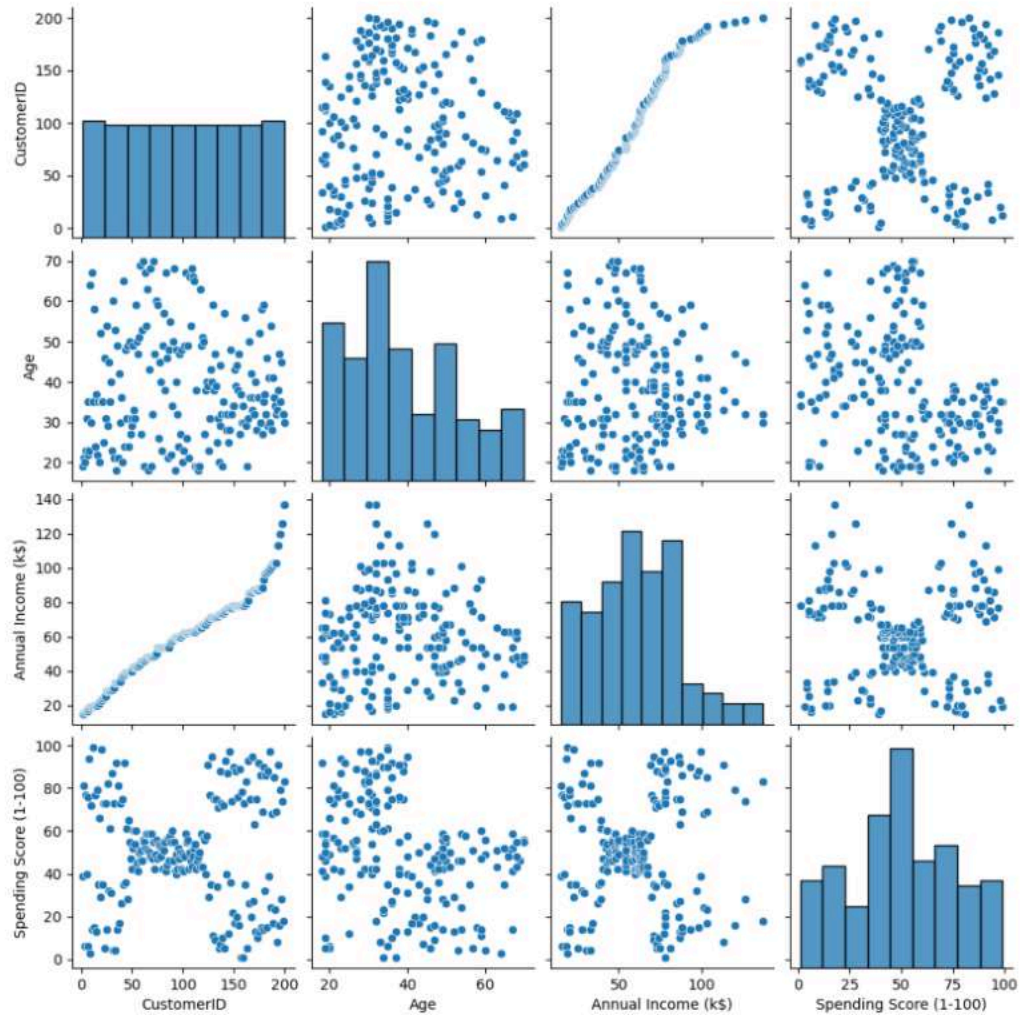
```
In [4]: df.head()
```

```
Out[4]:
```

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

```
In [5]: sns.pairplot(df)
```

```
Out[5]: <seaborn.axisgrid.PairGrid at 0x170e8e47850>
```



```
In [6]: features=df.iloc[:,[3,4]].values
```

```
In [7]: from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)
```

C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:870: FutureWarning: The default value of `n\_init` will change from 10 to 'auto' in 1.4. Set the value of `n\_init` explicitly to suppress the warning

C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\cluster_kmeans.py:1382: UserWarning: KMeans is known to have a memory leak on Windows with MKL, when there are less chunks than available threads. You can avoid it by setting the environment variable OMP_NUM_THREADS=1.

warnings.warn(

Out[7]: KMeans(n_clusters=5)

In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.

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```
In [8]: Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
Final.head()
```

C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_8116\470183701.py:2: SettingWithCopyWarning:

A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy (https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy)

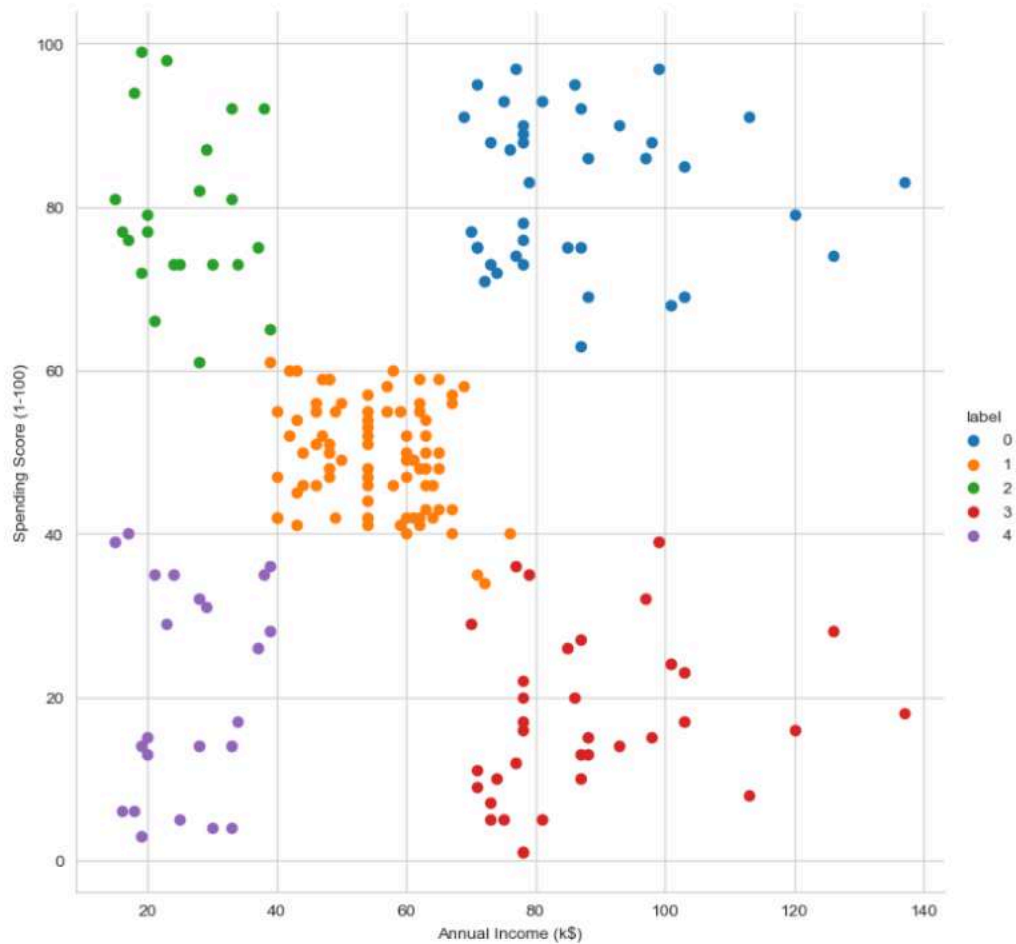
```
Final['label']=model.predict(features)
```

Out[8]:

	Annual Income (k\$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	2
2	16	6	4
3	16	77	2
4	17	40	4

	Annual Income (k\$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	2
2	16	6	4
3	16	77	2
4	17	40	4

```
In [9]: sns.set_style("whitegrid")
sns.FacetGrid(Final, hue="label", height=8) \
.map(plt.scatter, "Annual Income (k$)", "Spending Score (1-100)") \
.add_legend();
plt.show()
```



```
In [10]: features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
    model=KMeans(n_clusters=i)
    model.fit(features_el)
    wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```